Estimation and Inference on Two-Threshold Value Functionals[*]

Zhenxiao Chen[†]

July 27, 2026

Version 2

Abstract

We study estimation and inference for value functionals of the form $\Theta_0 = \mathbb{E}\big[\mathbb{1}\{g_0(X) \geq 0\}\, \mathbb{1}\{h_0(X) \geq 0\}\, v_0(X)\big]$, where g_0 and h_0 are two nonparametric contrast functions estimated from the same experiment. Such *two-threshold* functionals arise whenever a planner imposes a pointwise admissibility screen on top of a benefit margin: leading examples are the share of the population that benefits in the short run but is harmed in the long run under a surrogate-index treatment rule (Athey et al., 2024), and treatment-path welfare components of an optimal dynamic regime. We generalize the single-threshold theory of Chen et al. (2025): the pathwise derivative is a sum of *two* boundary (Hausdorff) integrals, the plug-in estimator converges at the one-dimensional nonparametric rate $n^{-s/(2s+1)}$ (attained under $s \geq d$), and a two-band sieve-Riesz variance estimator delivers valid studentized inference. Our main new result characterizes the *second-order* pathwise derivative: in addition to two margin curvature terms it contains a codimension-two *corner* integral over $\{g_0 = 0\} \cap \{h_0 = 0\}$, together with two corner *diagonal* terms weighted by the cosine of the intersection angle. The stochastic diagonal generated by the corner is of order K_n/n; the margin diagonals are bounded by $K_n^{1+1/d}/n$, so the corner is smaller by at most a factor $K_n^{-1/d}$ — but it is large enough to be rate-relevant whenever $s < d-1$, and unlike the margin bound it is not subject to the mesh cancellation we document. Consequently, margin-by-margin quadratic debiasing à la Chen and Gao (2026) leaves a corner cross term that forfeits the minimax rate on the window $(d+1)/2 < s \leq d-1$, nonempty for $d \geq 4$; we

[*] This section extends the single-threshold framework of Chen et al. (2025) and its debiased-estimation companion to value functionals defined by the intersection of two estimated decision boundaries. It is written to be self-contained; notation follows the JMP draft `may_2026.tex`.

[†] Department of Economics, University of Pennsylvania, USA, zxchen@upenn.edu.

construct corner-aware split-sample and leave-one-out debiased estimators that attain the minimax rate under the weaker smoothness $s > (d+1)/2$, and give a tuning-free half-sample jackknife representation of the split-sample correction. Finally, we combine the quadratic correction with a cross-fitted sieve-Riesz first-order correction into a *doubly debiased* estimator that simultaneously accommodates generic machine-learning first stages and weak smoothness — the combination posed as an open question in the single-threshold setting. Monte Carlo experiments calibrated to the Banerjee et al. (2015) graduation experiment, following the Wasserstein-GAN calibration protocol of Chen and Ritzwoller (2023), illustrate the corner mechanism and delineate the finite-sample limits of machine-learning first stages.

Revision note (v2). This version corrects the order accounting of Section 3.2, which in v1 assigned the margin and corner diagonals a common order K_n/n. The margin diagonals are bounded by $K_n^{1+1/d}/n$; the corner diagonals are K_n/n. The corrected accounting makes the smoothness requirement $s \geq d$ of Theorem 2.2 exactly the condition under which the margin bound is negligible, and it isolates $(d+1)/2 < s \leq d-1$ as the precise window on which corner-awareness is a rate requirement. Remark 3.3 is a countervailing caution that neither v1 nor its critique raised: the margin bound involves the *gradient* of the oscillating sieve variance function integrated over a surface, and we show analytically and numerically that it is attained only when the margin resonates with the sieve mesh; on a generic oblique geometry the margin diagonal falls back to K_n/n, the same order as the corner. The corner bound has no such cancellation and is the robust one. Section 3.2 also corrects the corner coefficient itself, which in v1 tracked only the ρ_{SY}-driven cross term and omitted two $\cos\omega$-weighted diagonal terms present in the plug-in bias even when $\rho_{SY} = 0$ — though, as (24) shows, those two terms *are* removed by margins-only debiasing, so v1 was right that the leftover is the ρ_{SY}-driven cross term. Lemma 4.2, Proposition 4.7, and the proof sketches of Theorems 4.5, 5.2 and 5.5 have been restated with the assumptions they actually require; two claims of v1's Remarks 5.3 and 5.4 were incorrect and are corrected in place. Theorem 3.1 itself is unchanged and is now verified numerically to ten significant digits in Appendix C. The Monte Carlo tables are reproduced unchanged from v1; Section 6.4 is new and states explicitly which of them do and do not support the asymptotic claims, and which implementation choices must be repaired before the next revision.

1 Setup and Examples

1.1 Data and estimands

The observed data are an i.i.d. sample $\{Z_i = (X_i, W_i, S_i, Y_i)\}_{i=1}^n$, where $X_i \in \mathcal{X} \subset \mathbb{R}^d$ are covariates with density f_0 on a bounded rectangular support, $W_i \in \{0,1\}$ is a binary treatment with $(S_i(0), S_i(1), Y_i(0), Y_i(1)) \perp W_i \mid X_i$ and overlap $p_0(x) := \mathbb{E}[W_i \mid X_i = x] \in (0,1)$, S_i is a short-run outcome (or surrogate), and Y_i is a long-run outcome. Define the arm means and contrasts

$$\mu_{S,0}(x,w) := \mathbb{E}[S_i \mid X_i = x, W_i = w], \qquad \tau_S(x) := \mu_{S,0}(x,1) - \mu_{S,0}(x,0), \tag{1}$$

$$\mu_{Y,0}(x,w) := \mathbb{E}[Y_i \mid X_i = x, W_i = w], \qquad \tau_Y(x) := \mu_{Y,0}(x,1) - \mu_{Y,0}(x,0). \tag{2}$$

Throughout we write $\gamma_0 := (g_0, h_0)$ for a generic pair of identified contrast functions built from $(\mu_{S,0}, \mu_{Y,0})$ — in the leading example $g_0 = \tau_S$ and $h_0 = -\tau_Y$ — and consider the *two-threshold value functional*

$$\Theta(\gamma_0) := \int_{\mathcal{X}} \mathbb{1}\{g_0(x) \geq 0\}\, \mathbb{1}\{h_0(x) \geq 0\}\, v_0(x)\, f(x)\, \mathrm{d}x, \tag{3}$$

where v_0 is a known, user-chosen value weight and f is the covariate density of the target population. As in the single-threshold analysis we distinguish the *known-density* case (f given, integral computed numerically) and the *empirical* case ($f = f_0$ unknown, integral replaced by the sample average $n^{-1} \sum_i \mathbb{1}\{\cdot\} v_0(X_i)$); the first-order asymptotic theory is identical in both, because the additional sampling error of the empirical measure is $O_p(n^{-1/2})$, which is negligible relative to the slower boundary-driven rate of (3). It is *not* negligible at the sample sizes of Section 6, and we therefore carry it explicitly in the variance formula (8) below.

1.2 Leading examples

Example 1.1 (Surrogate-induced harm share)**.** *A planner who observes only the short-run outcome adopts the surrogate-index rule "treat iff $\tau_S(x) \geq 0$." The long-run criterion would instead require $\tau_Y(x) \geq 0$. With $g_0 = \tau_S$, $h_0 = -\tau_Y$, and $v_0 \equiv 1$, (3) becomes*

$$\theta_0 = \mathbb{P}_f\big(\tau_S(X) \geq 0,\ \tau_Y(X) \leq 0\big),$$

the mass of the population that the surrogate rule treats but that is harmed in the long run — a direct diagnostic of the surrogate-index approach of Athey et al. (2024). This is the functional we take to the calibrated Monte Carlo of Section 6.

Example 1.2 (Admissibility-constrained policy value). *In the two-item-checklist planner problem of the JMP draft, a hard admissibility screen $g_0(x) \geq 0$ (e.g. no expected severe side effect) is imposed before the benefit margin $h_0(x) \geq 0$; treated-population aggregates take the form* (3) *with v_0 equal to a utility, cost, or eligibility weight.*

Example 1.3 (Dynamic treatment-path welfare). *The $(1,1)$ path-welfare component of an optimal two-stage dynamic regime, $V_{11}^* = \mathbb{E}[Y^{(1,1)}\mathbb{1}\{\kappa(S) \geq 0\}\mathbb{1}\{\delta(X^{(1)}) \geq 0\}]$, has the same two-threshold structure, with the additional complication that the first-stage contrast κ is itself a functional of the second-stage threshold. The static analysis of this section is the template for that case; the extra "moving-boundary-within-a-moving-boundary" terms are treated in the DTR section of the draft.*

1.3 Geometry and maintained assumptions

Write $w(x) := v_0(x)f(x)$ for the boundary weight, and define the two *margins* and the *corner*

$$\begin{aligned} \mathcal{M}_g &:= \{x : g_0(x) = 0,\ h_0(x) > 0\}, \qquad \mathcal{M}_h := \{x : h_0(x) = 0,\ g_0(x) > 0\}, \\ \mathcal{C} &:= \{x : g_0(x) = h_0(x) = 0\}. \end{aligned} \tag{4}$$

Assumption 1.4 (Model). *(a) $\{Z_i\}$ is i.i.d.; $\mathcal{X}$ is bounded rectangular; overlap holds; (b) $\mu_{S,0}(\cdot, w), \mu_{Y,0}(\cdot, w) \in \Lambda^s(\mathcal{X})$ (Hölder class) with $s > 1$, for $w \in \{0,1\}$; (c) the conditional second moments $\sigma_S^2(x,w), \sigma_Y^2(x,w)$ of the regression residuals $\epsilon_{S,i} := S_i - \mu_{S,0}(X_i, W_i)$, $\epsilon_{Y,i} := Y_i - \mu_{Y,0}(X_i, W_i)$ are bounded and bounded away from zero, the conditional correlation $\rho_{SY}(x,w) := \mathrm{Corr}(\epsilon_{S,i}, \epsilon_{Y,i} \mid X_i = x, W_i = w)$ is well defined, and $\mathbb{E}[|\epsilon_{o,i}|^{2+\varsigma} \mid X_i, W_i]$ is bounded for some $\varsigma > 0$ and $o \in \{S, Y\}$; (d) v_0 is bounded and continuously differentiable, and f is bounded with bounded gradient, absolutely continuous w.r.t. f_0 with bounded Radon–Nikodym derivative bounded away from zero on a neighborhood of $\mathcal{M}_g \cup \mathcal{M}_h$.*

Assumption 1.5 (Regular margins and transversal corner). *g_0 and h_0 are twice continuously differentiable in neighborhoods of $\{g_0 = 0\}$ and $\{h_0 = 0\}$ respectively, with (a) $\|\nabla g_0\| \geq \underline{c} > 0$ on $\{g_0 = 0\}$ and $\|\nabla h_0\| \geq \underline{c} > 0$ on $\{h_0 = 0\}$; (b) (*transversality*) on $\mathcal{C}$, the angle $\omega(x)$ between $\nabla g_0(x)$ and $\nabla h_0(x)$, defined by $\cos\omega(x) = \langle \nabla g_0, \nabla h_0 \rangle / (\|\nabla g_0\| \, \|\nabla h_0\|)$, satisfies $|\sin\omega(x)| \geq \underline{c}_T > 0$; (c) $\mathcal{H}^{d-1}(\mathcal{M}_g) + \mathcal{H}^{d-1}(\mathcal{M}_h) < \infty$ and $\mathcal{H}^{d-2}(\mathcal{C}) < \infty$; (d) (*no support edge*) the level sets $\{g_0 = 0\}$ and $\{h_0 = 0\}$ meet $\partial\mathcal{X}$ only where w vanishes, so that no margin terminates at the edge of the covariate support and the boundary calculus below involves only the margins and their common corner.*

Under Assumption 1.5, $\mathcal{M}_g$ and $\mathcal{M}_h$ are $(d-1)$-dimensional C^2 submanifolds-with-boundary whose common relative boundary is the $(d-2)$-dimensional corner $\mathcal{C}$ (a finite set of points when $d = 2$; empty when $d = 1$ or when the constraints never bind jointly). Transversality rules out tangential intersections, at which the corner would inflate to a higher-dimensional set and the second-order calculus below would fail.

Remark 1.6 (Assumption 1.5(d) is substantive). *Condition (d) is not a technicality. If a margin exits through $\partial\mathcal{X}$ at a point where $w > 0$, then $\mathcal{M}_g$ has a second kind of relative boundary —* a support edge *— and the second derivative* (10) *acquires an extra $\mathcal{H}^{d-2}$ integral over $\{g_0 = 0\} \cap \partial\mathcal{X}$. Because the support is* fixed *while the level set flows, only the "induced motion of the cut" mechanism (step (4) of Appendix A) operates: the edge contributes a u^2* diagonal *term*

$$-\int_{\{g_0=0\}\cap\partial\mathcal{X}} \frac{u^2\, w\, \cos\omega_\partial}{\|\nabla g_0\|^2\, |\sin\omega_\partial|}\, \mathrm{d}\mathcal{H}^{d-2}, \qquad \cos\omega_\partial := \frac{\langle \nabla g_0, \nu\rangle}{\|\nabla g_0\|},$$

with ν the outward unit normal of $\mathcal{X}$, and no *cross term (there is no second perturbed surface to cross with). It vanishes when the margin meets $\partial\mathcal{X}$ orthogonally. The first derivative* (5) *is unaffected. Appendix C, item 6, verifies this display to ten digits. We flag it because the principal calibrated design of Section 6 violates (d): see Section 6.4, item (L3), which reports twelve such crossings.*

2 First-Order Theory

2.1 Pathwise derivative: two boundary integrals

For directions (u, v) (perturbations of (g_0, h_0) induced by perturbations of the four arm means), let $\gamma_t := (g_0 + tu,\ h_0 + tv)$.

Proposition 2.1 (First derivative). *Under Assumptions 1.4–1.5,*

$$D\Theta(\gamma_0)[u, v] := \frac{\mathrm{d}}{\mathrm{d}t}\Theta(\gamma_t)\Big|_{t=0} = \int_{\mathcal{M}_g} \frac{u(x)\, w(x)}{\|\nabla g_0\|}\, \mathrm{d}\mathcal{H}^{d-1}(x) + \int_{\mathcal{M}_h} \frac{v(x)\, w(x)}{\|\nabla h_0\|}\, \mathrm{d}\mathcal{H}^{d-1}(x). \quad (5)$$

Proof sketch. Apply the generalized Leibniz rule (Reynolds transport theorem; Delfour and Zolésio, 2001, Thm. 4.2) to the moving region $\Omega_t = \{g_t \geq 0\} \cap \{h_t \geq 0\}$ with fixed integrand w. The boundary $\partial\Omega_0$ decomposes, up to the $\mathcal{H}^{d-1}$-null corner, into the two faces $\mathcal{M}_g$ and $\mathcal{M}_h$; on each face the normal velocity is that of the corresponding level set, $-u/\|\nabla g_0\|$ and $-v/\|\nabla h_0\|$ respectively, and the coarea representation of the surface measure yields (5). $\square$

Three features of (5) drive everything that follows.

(i) Irregularity. Each term is a $(d-1)$-dimensional Hausdorff integral: a bounded linear functional of (u, v) on C^1 but *not* on $L^2(f)$. Hence, exactly as in the single-threshold case, no L^2-Riesz representer (influence function) exists whenever w does not vanish on $\mathcal{M}_g \cup \mathcal{M}_h$, and Θ_0 is not $\sqrt{n}$-estimable. The welfare-type exception — boundary weights that vanish on their own margins, as for total dynamic welfare V^* in the DTR section — is the regular special case.

(ii) Two bands. The derivative aggregates information about *two different contrasts* on *two different thin sets*. The variance of a plug-in estimator therefore has two band contributions, and, because $\hat{g}$ and $\hat{h}$ are estimated from the same observations, a within-unit covariance term appears whenever $\rho_{SY} \neq 0$.

(iii) The corner is invisible at first order. $\mathcal{C}$ has $\mathcal{H}^{d-1}$-measure zero and does not appear in (5); it enters only the second-order expansion (Section 3), which is precisely why it has been ignorable in first-order analyses and why it becomes relevant for debiasing.

2.2 Plug-in estimator, rate, and minimax optimality

Let $\hat{\mu}_S, \hat{\mu}_Y$ be stacked sieve least-squares estimators of the arm means on tensor-product B-spline bases of dimension K_n (per arm), let $(\hat{g}, \hat{h})$ be the implied contrast estimates, and let

$$\hat{\Theta} := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{\hat{g}(X_i) \geq 0\} \, \mathbb{1}\{\hat{h}(X_i) \geq 0\} \, v_0(X_i)$$

be the (empirical-measure) plug-in estimator; in the known-density case the sample average is replaced by numerical integration against f.

Theorem 2.2 (Plug-in: rate and asymptotic normality)**.** *Let Assumptions 1.4–1.5 hold together with the sieve regularity conditions of Chen et al. (2025, Thm. 3) applied to each of the four arm regressions, and let $K_n \asymp n^{d/(2s+1)}$ up to logarithmic factors. Then*

$$\frac{\sqrt{n}\,(\hat{\Theta} - \Theta_0)}{\sigma_n} \xrightarrow{d} \mathcal{N}(0,1), \qquad \sigma_n^2 := \left\| v_{K_n}^* \right\|_{sd}^2 \asymp K_n^{1/d},$$

where $v_{K_n}^*$ is the sieve-Riesz representer of the two-band derivative (5) defined in (6) below. In particular $\hat{\Theta} - \Theta_0 = O_p(n^{-s/(2s+1)})$ provided $s \geq d$. Moreover, $n^{-s/(2s+1)}$ is a minimax lower-bound rate for estimating Θ_0 over Hölder balls Λ_c^s.

Proof sketch. Upper bound and CLT. The expansion $\hat{\Theta} - \Theta_0 = D\Theta[\hat{\gamma} - \gamma_0] + \frac{1}{2} D^2\Theta[\hat{\gamma} - \gamma_0, \hat{\gamma} - \gamma_0] + R_3$ has a linear term that is a sum of two submanifold integrals of sieve LS errors; each is handled by the single-threshold argument (Chen and Gao, 2026, Thm. 8; Chen et al., 2025, Thm. 3) with the truncation indicators $\mathbb{1}\{h_0 > 0\}$, $\mathbb{1}\{g_0 > 0\}$ carried through as fixed bounded weights, and the two contributions are jointly asymptotically normal by the Cramér–Wold device applied to the stacked representer. The quadratic term is controlled in Section 3.2: its dominant piece is the margin stochastic diagonal, of order $K_n^{1+1/d}/n$, and Table 1 shows that this is $O(r_n^*)$ at $K_n \asymp n^{d/(2s+1)}$ *exactly* when $s \geq d$, which is the source of the smoothness requirement. *Lower bound (nesting).* Consider the subclass of models in which $h_0 \geq \underline{\eta} > 0$ on the support, so that $\mathbb{1}\{h_0 \geq 0\} \equiv 1$ and Θ reduces to the single-threshold value functional $V(g_0)$; every estimator of Θ_0 is then an estimator of $V(g_0)$, and the minimax lower bound of Chen and Gao (2026, Thm. 5) / Chen et al. (2025, Thm. 5) applies verbatim. Hence the two-threshold problem is at least as hard, and the plug-in rate is sharp. □

2.3 Two-band sieve-Riesz variance

Let $\psi(x, w)$ denote the stacked per-arm sieve regressors of one outcome equation and $b(x) := (\psi^{(K_1)}(x)', -\psi^{(K_0)}(x)')'$ the contrast basis. The (infeasible) sieve-Riesz representer of (5) restricted to the sieve space is, for each outcome equation $o \in \{S, Y\}$,

$$v^*_{K_n,o}(x, w) := \psi(x, w)' R^{-1}_{K_n,o} D_o\Theta(\gamma_0)[\,\psi\,], \qquad R_{K_n,o} := \mathbb{E}[\psi(X_i, W_i)\psi(X_i, W_i)'], \tag{6}$$

where $D_o\Theta[\psi]$ stacks the coordinatewise boundary integrals of (5) over the relevant margin. Feasible versions replace population by empirical moments and the Hausdorff integrals by ε-band sample averages. Because an ε-band average of a function φ over $\{|\hat{g}| < \varepsilon\}$ converges by the coarea formula to $\int_{\{g_0=0\}} \varphi f_0/\,\|\nabla g_0\|\;\mathrm{d}\mathcal{H}^{d-1}$ — against the *sampling* density f_0, not against $w = v_0 f$ — the band average must be reweighted by w/f_0:

$$\begin{aligned}
\widehat{D_g\Theta}[\,b\,] &= \frac{+1}{2\varepsilon_S}\,\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{|\hat{g}(X_i)| < \varepsilon_S,\ \hat{h}(X_i) > 0\}\,\frac{v_0(X_i)f(X_i)}{\hat{f}_0(X_i)}\,b(X_i), \\
\widehat{D_h\Theta}[\,b\,] &= \frac{-1}{2\varepsilon_Y}\,\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{|\hat{\tau}_Y(X_i)| < \varepsilon_Y,\ \hat{g}(X_i) \geq 0\}\,\frac{v_0(X_i)f(X_i)}{\hat{f}_0(X_i)}\,b(X_i).
\end{aligned} \tag{7}$$

In the empirical-measure case $f = f_0$ and the weight reduces to $v_0(X_i)$; in Example 1.1, where $v_0 \equiv 1$, it reduces to 1, which is the case implemented in Section 6. (v1 stated (7) without the weight, i.e. only for that case.) The displays are instantiated for the harm-share convention of Example 1.1: for a generic pair (g, h) both bands carry a $+$ sign in (g, h) coordinates, and the $-$ sign of the second display is the Jacobian of $h_0 = -\tau_Y$ (raising τ_Y shrinks the harmed region).

The variance estimator sums the *per-unit* influence contributions of the two outcome equations *before* squaring, and adds the empirical-measure term:

$$\widehat{\mathrm{Var}}(\hat{\Theta}) = \frac{\hat{\sigma}_n^2}{n} + \frac{\hat{\varsigma}^2}{n}, \quad \hat{\sigma}_n^2 := \frac{1}{n}\sum_{i=1}^{n}\left(\widehat{v_S^*}(X_i, W_i)\,\hat{\epsilon}_{S,i} + \widehat{v_Y^*}(X_i, W_i)\,\hat{\epsilon}_{Y,i}\right)^2, \tag{8}$$

where $\hat{\varsigma}^2 := n^{-1}\sum_i \left(\mathbb{1}\{\hat{g}_i \geq 0\}\mathbb{1}\{\hat{h}_i \geq 0\}v_0(X_i) - \hat{\Theta}\right)^2$ reduces to $\hat{\Theta}(1-\hat{\Theta})$ when $v_0 \equiv 1$. Summing the two influence contributions before squaring means the within-unit covariance of the short- and long-run residuals ($\rho_{SY} \neq 0$) is captured automatically. Here $\|v\|_{sd}^2 := \mathbb{E}[v(X_i, W_i)^2\sigma^2(X_i, W_i)]$ denotes the standard-deviation norm and $v_{K_n}^*$ the stacked (S, Y) representer. Adding the two components of (8) ignores their covariance — the same X_i enter the empirical measure and the first-stage residuals — which is of smaller order than either and which we do not estimate. Because $\sigma_n^2 \asymp K_n^{1/d} \to \infty$ while $\hat{\varsigma}^2 = O(1)$, the second term of (8) is asymptotically negligible; it is 6–12% of the reported standard error at the sample sizes of Section 6, and omitting it produces a spurious SE/SD ratio of 0.87–0.95.

Proposition 2.3 (Ratio consistency)**.** *Under the conditions of Theorem 2.2 and $\varepsilon_n \downarrow 0$ with $\varepsilon_n^{-1}(\|\hat{g} - g_0\|_\infty + \left\|\hat{h} - h_0\right\|_\infty) = o_p(1)$ and $n\varepsilon_n/K_n \to \infty$, $\hat{\sigma}_n^2/\sigma_n^2 \xrightarrow{p} 1$, and the studentized statistic $\sqrt{n}(\hat{\Theta} - \Theta_0)/\hat{\sigma}_n$ is asymptotically standard normal.*

Ratio consistency is the relevant notion because $\sigma_n^2 \to \infty$ in the irregular regime. The proof adapts the single-threshold argument to each band, using Evans and Gariepy (2015, Thm. 3.13(iii)) for the ε-band-to-Hausdorff convergence on each truncated margin. The single-threshold statement requires only $n\varepsilon_n \to \infty$; we impose the stronger $n\varepsilon_n/K_n \to \infty$ because (7) estimates a K_n-vector from the $\asymp n\varepsilon_n$ observations that fall in the band, and the feasible representer inverts a $K_n \times K_n$ Gram matrix against it. This is a sufficient-side strengthening, not an imported result. The requirement $\varepsilon_n \to 0$ is essential: a bandwidth that stabilizes at a fixed positive fraction of $\mathrm{sd}(\hat{\tau})$ estimates a smeared functional, not $D\Theta$, and Proposition 2.3 does not apply to it. See Section 6.4, item (L5).

3 Second-Order Theory: the Corner

3.1 The second pathwise derivative

The plug-in expansion

$$\hat{\Theta} - \Theta_0 = D\Theta(\gamma_0)[\hat{\gamma} - \gamma_0] + \tfrac{1}{2}D^2\Theta(\gamma_0)[\hat{\gamma} - \gamma_0, \hat{\gamma} - \gamma_0] + R_3, \tag{9}$$

with R_3 a third-order remainder, is what limits the smoothness requirement of Theorem 2.2. The new object is $D^2\Theta$.

Theorem 3.1 (Second derivative: two margins and a corner)**.** *Under Assumptions 1.4–1.5, with $n_g := \nabla g_0 / \|\nabla g_0\|$, $H_g := \operatorname{div}(n_g)$ the mean curvature of $\{g_0 = 0\}$ (and symmetrically for h), Θ is twice pathwise differentiable with*

$$D^2\Theta(\gamma_0)[(u,v),(u,v)] = Q_g[u,u] + Q_h[v,v] + 2\int_{\mathcal{C}} \frac{u(x)\,v(x)\,w(x)}{\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega(x)|}\,\mathrm{d}\mathcal{H}^{d-2}(x), \quad (10)$$

where the margin quadratic forms are

$$\begin{aligned} Q_g[u,u] = &-\int_{\mathcal{M}_g} \left[\frac{u\,(n_g\!\cdot\!\nabla u)\,w}{\|\nabla g_0\|^2} + \frac{u}{\|\nabla g_0\|}\left\{ n_g\!\cdot\!\nabla\Big(\frac{u\,w}{\|\nabla g_0\|}\Big) + \frac{u\,w}{\|\nabla g_0\|}\,H_g \right\}\right] \mathrm{d}\mathcal{H}^{d-1} \\ &- \int_{\mathcal{C}} \frac{u^2(x)\,w(x)\,\cos\omega(x)}{\|\nabla g_0\|^2\,|\sin\omega(x)|}\,\mathrm{d}\mathcal{H}^{d-2}(x), \end{aligned} \quad (11)$$

and Q_h is the mirror image (swap $g \leftrightarrow h$, $u \leftrightarrow v$, and restrict to $\mathcal{M}_h$).

Proof sketch; full derivation in Appendix A. Differentiate each term of (5) along γ_t. Three effects arise for the $\mathcal{M}_g$-term: (i) the integrand $u\,w/\|\nabla g_t\|$ changes (the $n_g\!\cdot\!\nabla u$ term); (ii) the level set $\{g_t = 0\}$ flows with normal velocity $-u/\|\nabla g_0\|$, producing the tangential-gradient and curvature terms of the moving-level-set lemma of Chen and Gao (2026, Lem. PathD–Level) applied with weight $u\,w/\|\nabla g_0\|$; (iii) the *truncation* $\{h_t > 0\}$ of $\mathcal{M}_g$ moves. Effect (iii) has two parts: the direct motion of the cut in the v-direction, an edge integral over $\mathcal{C}$ with within-manifold coarea factor $\|P_T\nabla h_0\| = \|\nabla h_0\|\,|\sin\omega|$ (this is the cross term, and it appears once from each margin, giving the factor 2 in (10)); and the induced motion of the edge caused by the flow of the manifold itself across the fixed cut — pulling the cut condition back through the normal flow perturbs it by $-tu\,\|\nabla h_0\|\cos\omega/\|\nabla g_0\|$, which yields the diagonal corner term in (11). Transversality (Assumption 1.5(b)) keeps all corner factors bounded. □

When $d = 2$ the corner is a finite set of points and $\mathcal{H}^0$ is counting measure: the cross term is $2\sum_{x\in\mathcal{C}} u(x)v(x)w(x)/(\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|)(x)$.

Remark 3.2 (Numerical verification)**.** *Theorem 3.1 is verified to ten significant digits in Appendix C on closed-form $d = 2$ designs that between them activate every term of (10)–(11): a radially weighted disk (the transport and curvature terms of Q_g); a disk with a non-constant direction u (the $u\,n_g\!\cdot\!\nabla u$ term); a disk cut by an oblique chord and the intersection of two disks (both corner diagonals and the corner cross term, at unequal u, v, at one-sided perturbations,* *and at both signs of* $\cos\omega$*). The same appendix verifies the polarization identity* (24) *and the support-edge term of Remark 1.6. This is the part of the analysis that is checked rather than asserted.*

3.2 Orders of the second-order terms

Write $\Delta_g := \hat{g} - g_0$, $\Delta_h := \hat{h} - h_0$ for sieve LS first stages with K_n basis functions per arm. Decompose each error into an approximation bias and a linear stochastic part,

$$\Delta_g = b_g + u_g, \qquad u_g(x) := \frac{1}{n}\sum_i s_i(x)\,\epsilon_{S,i}, \qquad s_i(x) = b(x)'G^{-1}b(X_i), \tag{12}$$

with $\|b_g\|_\infty \lesssim K_n^{-s/d}$, $\|\nabla b_g\|_\infty \lesssim K_n^{-(s-1)/d}$, $G := \mathbb{E}[b(X_i)b(X_i)']$, and similarly for Δ_h with residuals $\epsilon_{Y,i}$. Note that the sieve influence weights s_i are *common* to the two regressions, because the same design matrix and the same basis are used; this is what generates the within-unit coupling below.

Two scaling facts for tensor-product B-splines with mesh $\varkappa = K_n^{-1/d}$ drive the whole accounting:

$$b(x)'G^{-1}b(x) \asymp K_n, \qquad b(x)'G^{-1}\nabla b(x) \asymp K_n^{1+1/d} \tag{13}$$

(each derivative of a basis function costs a factor $\varkappa^{-1} = K_n^{1/d}$). Consequently, pointwise,

$$\begin{aligned} \mathbb{E}[u_g(x)^2] &\asymp \frac{\overline{\sigma_S^2}(x)}{n}\,K_n, \qquad\qquad \mathbb{E}[u_g(x)\,(n_g\cdot\nabla u_g)(x)] \asymp \frac{\overline{\sigma_S^2}(x)}{n}\,K_n^{1+1/d},\\ \mathbb{E}[u_g(x)u_h(x)] &\asymp \frac{\overline{\rho\sigma_S\sigma_Y}(x)}{n}\,K_n, \end{aligned} \tag{14}$$

where the arm-aggregated conditional second moments of the contrast regressions are

$$\overline{\sigma_S^2}(x) := \sum_{w\in\{0,1\}} \frac{\sigma_S^2(x,w)}{\mathbb{P}(W=w \mid X=x)}, \qquad \overline{\rho\sigma_S\sigma_Y}(x) := \sum_{w\in\{0,1\}} \frac{\rho_{SY}(x,w)\,\sigma_S(x,w)\,\sigma_Y(x,w)}{\mathbb{P}(W=w \mid X=x)}. \tag{15}$$

(v1 wrote the third display of (14) with an undefined normalization and with "$\asymp$" applied to a signed quantity. The correct statement is that $\mathbb{E}[u_g(x)u_h(x)]$ equals $\overline{\rho\sigma_S\sigma_Y}(x)\,b(x)'G^{-1}b(x)/n$ exactly; its *magnitude* is of order $|\overline{\rho\sigma_S\sigma_Y}(x)|\,K_n/n$, and its sign is that of ρ_{SY}.)

Taking expectations term by term in (10) gives the *upper bounds* in Table 1. The essential point — and the correction to v1 — is that the margin forms Q_g, Q_h contain a factor ∇u while the corner forms do not. *The margin diagonals are therefore bounded by* $K_n^{1/d}$ *times the corner diagonals*, not by the same quantity. Every entry of Table 1 is an $O(\cdot)$, and this is all the theorems below use; Remark 3.3 discusses which entries are actually attained.

Table 1: Upper bounds on the second-order terms and the smoothness needed for each to be $O(r_n^*)$, $r_n^* = n^{-s/(2s+1)}$, at $K_n \asymp n^{d/(2s+1)}$. "Diagonal" terms enter the estimator error linearly and must be compared with r_n^* itself; "off-diagonal" (cross-half / $i \neq j$) terms are mean-zero degenerate U-statistics. Derivations of the off-diagonal orders are in Appendix B. Entries are bounds, not exact orders: see Remark 3.3.

Component	Term	Order	$O(r_n^*)$ iff
Stochastic diagonals (removed by SS/LOO debiasing)			
Margin	$Q_g[u_g, u_g]$, $Q_h[u_h, u_h]$	$K_n^{-1+1/d}/n$	$s \geq d$
Corner	all three corner pieces	K_n/n	$s \geq d-1$
Approximation-bias diagonals (removed by undersmoothing)			
Margin	$Q_g[b_g, b_g]$	$K_n^{-(2s-1)/d}$	$s > 1$
Corner	$C[b_g, b_h]$ etc.	$K_n^{-2s/d} = r_n^{*2}$	always
Bias $\times$ stochastic cross terms (removed by neither)			
Margin	$Q_g[b_g, u_g]$	$K_n^{(1-s)/d}\sqrt{K_n/n}$	$s \geq (d+1)/2$
Corner	$C[b_g, u_h]$	$K_n^{-s/d}\sqrt{K_n/n}$	$s \geq (d-1)/2$
Cross-half off-diagonals (mean zero; degenerate)			
Margin	gradient part	$n^{-1}K_n^{(d+3)/(2d)}$	$s \geq (d+1)/2$
Corner	—	$n^{-1}K_n^{(d+2)/(2d)}$	$s \geq d/2$
Margin	curvature part	$n^{-1}K_n^{(d+1)/(2d)}$	$s \geq (d-1)/2$

Three consequences follow, and they replace the corresponding discussion of v1.

(a) The plug-in's smoothness requirement comes from the margin diagonal. $K_n^{1+1/d}/n = n^{-(2s-d)/(2s+1)}$ is $O(r_n^*)$ if and only if $s \ge d$ — precisely the condition in Theorem 2.2. In v1 this requirement was attributed to K_n/n, which would have given $s \ge d-1$, and the one-derivative discrepancy was left unexplained. With the bound of Table 1 the condition $s \ge d$ is exactly what is needed for the bound to be negligible. We do *not* claim it is necessary; see Remark 3.3.

(b) The corner is no larger than the margins and is still rate-relevant on a nonempty window. $K_n/n = n^{-(2s+1-d)/(2s+1)}$ is $O(r_n^*)$ if and only if $s \ge d-1$. Since split-sample debiasing (Section 4) extends the guarantee down to $s > (d+1)/2$, the corner diagonal is rate-relevant — i.e. larger than r_n^* inside the regime where the theory is supposed to deliver — exactly on

$$\tfrac{d+1}{2} < s \le d-1, \qquad \text{nonempty iff } d \ge 4. \tag{16}$$

For $d \in \{2,3\}$ the window is empty, so at the $d = 2$ of our designs corner-awareness is a *finite-sample constants* question, not an asymptotic rate question. This is a weaker claim than v1's abstract made, and it is the correct one.

(c) The corner coefficient has three pieces, not one. Collecting the corner terms of (10)–(11) and using (14), the corner's contribution to $\mathbb{E}[\frac{1}{2}D^2\Theta[\Delta,\Delta]]$ is

$$\mathcal{B}_{\mathcal{C}} = \int_{\mathcal{C}} \frac{w(x)}{2\,|\sin\omega(x)|}\left[\frac{2V_{gh}(x)}{\|\nabla g_0\|\,\|\nabla h_0\|} - \cos\omega(x)\left(\frac{V_g(x)}{\|\nabla g_0\|^2} + \frac{V_h(x)}{\|\nabla h_0\|^2}\right)\right] \mathrm{d}\mathcal{H}^{d-2}(x), \tag{17}$$

where $V_g := \mathbb{E}[\Delta_g^2]$, $V_h := \mathbb{E}[\Delta_h^2]$, $V_{gh} := \mathbb{E}[\Delta_g\Delta_h]$ evaluated at x. All three are $\asymp K_n/n$ up to their bias parts, so $\mathcal{B}_{\mathcal{C}} \asymp K_n/n$. Two readings matter. First, the bracket is a *difference*: the within-unit correlation term and the two $\cos\omega$-weighted own-variance terms can partially cancel, and the sign of the corner bias is not determined by $\operatorname{sign}(\rho_{SY})$ alone. Second, the corner contribution to the *plug-in* bias does not vanish when $\rho_{SY} = 0$: the two $\cos\omega$ terms survive, and they vanish only when the margins meet orthogonally. In the notation of (17), v1 tracked only the first term of the bracket. Note carefully, however, that the two $\cos\omega$ terms belong to Q_g and Q_h and are therefore removed by margin-by-margin debiasing; only the V_{gh} term survives it, and *that* term does vanish at $\rho_{SY} = 0$. Section 4 makes this precise (Proposition 4.7), and the two statements should not be conflated: (17) is the corner's contribution to the plug-in's bias, not the part that margins-only debiasing leaves behind.

Remark 3.3 (Are these bounds attained? A caution about mesh resonance)**.** *Table 1 reports bounds, and for the margin diagonal the bound need not be tight. Under conditional* *homoskedasticity the leading margin diagonal is, exactly,*

$$\mathbb{E}\big[Q_g[u_g,u_g]\big]_{grad} = -\frac{\sigma^2}{n}\int_{\mathcal{M}_g} \frac{w}{\|\nabla g_0\|^2}\, n_g\cdot\nabla\mathcal{V}\,\mathrm{d}\mathcal{H}^{d-1}, \qquad \mathcal{V}(x) := b(x)'G^{-1}b(x), \tag{18}$$

because $\mathbb{E}[u_g\,\partial_n u_g] = (\sigma^2/2n)\,\partial_n\mathcal{V}$*. Now* $\mathcal{V}\asymp K_n$ oscillates *at the mesh scale* $\varkappa = K_n^{-1/d}$*, so* $\|\nabla\mathcal{V}\|_\infty \asymp K_n^{1+1/d}$ *pointwise — but* (18) *integrates it against a smooth weight over a* $(d-1)$*-dimensional surface, and an oscillating integrand can cancel. Whether it does depends on the arithmetic relationship between the margin's normal direction and the tensor mesh. Writing* $\mathcal{V}$ *as a Fourier series in the mesh period, the surface integral retains a non-decaying contribution only from* resonant *frequency pairs, which requires the margin's slope to be rational with a small denominator. Numerically (Appendix C, item 5), with periodic cubic tensor B-splines in* $d=2$*: a mesh-aligned margin gives* $\asymp K_n^{1+1/d}$*, the exact diagonal* $\alpha=1$ *gives the same order with a smaller constant, and generic irrational slopes* ($1/\pi$*,* $\sqrt{2}$*,* $\sqrt{5}-1$) *give* $\asymp K_n$ *— a hundredfold gap at* $K_n^{1/2}=128$*.*

Three implications. (i) $s\ge d$ *in Theorem 2.2 is sufficient but we do not claim it necessary; on a generic geometry* $s\ge d-1$ *may suffice.* (ii) *No such mechanism applies to the corner:* V_g, V_h, V_{gh} *in* (17) *involve* $\mathcal{V}$ undifferentiated*, are of one sign, and are evaluated pointwise on a* $(d-2)$*-set, so* $\mathcal{B}_{\mathcal{C}}\asymp K_n/n$ *with no cancellation. The corner bound is the robust one.* (iii) *Consequently the safe statement of the margin-versus-corner comparison is an inequality, not an equality of ratios: the corner is never more than* $K_n^{1/d}$ *smaller than the margins, and on a generic oblique geometry the two are plausibly of the same order — which is where v1's instinct, though not its argument, pointed. Two caveats: the identity* $\mathbb{E}[u\,\partial_n u] = (\sigma^2/2n)\partial_n\mathcal{V}$ *requires homoskedasticity (otherwise the relevant object is* $n^{-1}b'G^{-1}\Omega G^{-1}\nabla b$*, not an exact gradient), and the evidence is a* $d=2$ *probe with straight margins. Establishing the exact order of* (18) *for curved margins and general* d *is open; we flag it because it is the one place where the accounting above could be conservative rather than wrong.*

A margin-by-margin application of the single-threshold correction of Chen and Gao (2026) removes the margin diagonals *and* the two $\cos\omega$ corner diagonals, but leaves the corner cross term $\mathcal{B}_{\mathcal{C}}^{\times}$ in place; Proposition 4.7 below records the implication.

4 Quadratic Debiasing

4.1 Split-sample estimator and a tuning-free representation

Randomly split the sample into halves A and B (stratified by W), run the stacked sieve regressions on each half, and let $\hat\gamma^A = (\hat g^A, \hat h^A)$, $\hat\gamma^B$ denote the two estimated pairs, $\bar\gamma :=$

$(\hat\gamma^A+\hat\gamma^B)/2$, and $\Delta := \hat\gamma^A - \hat\gamma^B$. The split-sample (SS) debiased estimator is

$$\hat\Theta_{SS} := \Theta(\bar\gamma) - \tfrac{1}{8}\, D^2\Theta(\bar\gamma)[\Delta,\Delta]. \tag{19}$$

The construction works because the approximation bias is common to the two halves and cancels from Δ: writing $\hat\gamma^j = \gamma_0 + b + u^j$ with a common deterministic b and independent mean-zero u^A, u^B,

$$\bar\Delta := \bar\gamma - \gamma_0 = b + \bar u, \quad \bar u = \tfrac12(u^A + u^B), \qquad \Delta = u^A - u^B, \tag{20}$$

and by bilinearity $D^2\Theta[\bar u,\bar u] - \frac14 D^2\Theta[\Delta,\Delta] = D^2\Theta[u^A,u^B]$, a mean-zero cross-half term. Thus (19) removes the *stochastic* diagonals — margins and corner alike — and nothing else. It does not touch the bias diagonals or the bias $\times$ stochastic cross terms; those are the third and fourth blocks of Table 1 and are controlled by smoothness, not by debiasing.

Assumption 4.1 (Path regularity)**.** *There is a neighborhood $\mathcal N$ of γ_0 in $C^3(\mathcal X)^2$ such that, for every $\gamma \in \mathcal N$, the level sets $\{g=0\}$, $\{h=0\}$ satisfy Assumption 1.5(a)–(d) with the same constants $\underline{c}, \underline{c}_T$; $w \in C^3$ in a neighborhood of $\{g_0 = 0\} \cup \{h_0 = 0\}$; and the map $t \mapsto \Theta(\gamma + t\eta)$ is four times continuously differentiable on $[-1,1]$ for $\gamma, \gamma+\eta \in \mathcal N$, with* $\sup_{|t|\le 1} \left| \frac{\mathrm d^4}{\mathrm dt^4}\Theta(\gamma+t\eta)\right| \lesssim \|\eta\|_{C^1}^4$.

Assumption 4.1 is what a fourth-order finite-difference argument needs and what Assumption 1.5 alone does not supply: Assumption 1.5 gives C^2 boundaries and Assumption 1.4(d) a C^1 weight, which suffice for (10) but not for two further derivatives along the path. It also requires *uniform* regularity of the estimated geometry along the whole segment $\bar\gamma + t\Delta$, $|t| \le \frac12$, not merely at γ_0; with sieve first stages this holds on an event of probability tending to one whenever $\|\hat\gamma^j - \gamma_0\|_{C^1} = o_p(1)$, which is $K_n^{1/2+1/d} n^{-1/2} + K_n^{-(s-1)/d} = o(1)$.

Lemma 4.2 (Half-sample jackknife representation)**.** *Under Assumption 4.1, approximating $D^2\Theta(\bar\gamma)[\Delta,\Delta]$ by the symmetric second difference at step $\delta = \frac12$ — i.e. evaluating Θ at $\bar\gamma \pm \frac12\Delta = \hat\gamma^A, \hat\gamma^B$ — gives the closed form*

$$\hat\Theta_{SS}^{\mathrm{jack}} = 2\,\Theta(\bar\gamma) - \tfrac12\big(\Theta(\hat\gamma^A) + \Theta(\hat\gamma^B)\big). \tag{21}$$

Writing $\phi(t) := \Theta(\bar\gamma + t\Delta)$, the approximation error is exactly

$$\hat\Theta_{SS}^{\mathrm{jack}} - \hat\Theta_{SS} = -\tfrac{1}{384}\,\phi^{(4)}(\xi) \quad \textit{for some } |\xi| \le \tfrac12.$$

Under the fourth-derivative bound of Assumption 4.1, $\phi^{(4)}(\xi) = O_p(\|\Delta\|_{C^1}^4) = O_p(K_n^{2+4/d} n^{-2})$, *which is $o(r_n^*)$ at $K_n \asymp n^{d/(2s+1)}$ if and only if*

$$s > \tfrac{2(d+1)}{3}. \tag{22}$$

Moreover (21) is exact whenever $t \mapsto \Theta(\bar{\gamma} + t\Delta)$ is a quadratic polynomial.

Proof. $\phi(\frac{1}{2}) + \phi(-\frac{1}{2}) - 2\phi(0) = \frac{1}{4}\phi''(0) + \frac{1}{192}\phi^{(4)}(\xi)$ by Taylor's theorem with Lagrange remainder, valid because $\phi \in C^4[-\frac{1}{2}, \frac{1}{2}]$ under Assumption 4.1. Substituting $D^2\Theta(\bar{\gamma})[\Delta, \Delta] = \phi''(0)$ into (19) and rearranging gives (21) with the stated error. The rate statement substitutes $\|\Delta\|_{C^1} \asymp K_n^{1/2+1/d} n^{-1/2}$ and $K_n \asymp n^{d/(2s+1)}$, giving exponent $(2d + 2 - 3s)/(2s + 1)$ relative to r_n^*. $\square$

Remark 4.3 (An open gap in the jackknife implementation)**.** *Condition (22) is* strictly stronger *than the $s > (d + 1)/2$ of Theorem 4.5: $2(d + 1)/3 > (d + 1)/2$ for every d, e.g. $s > 2$ against $s > 1.5$ at $d = 2$. So with the bound currently available, the tuning-free implementation (21) is not proved to inherit the weak-smoothness guarantee of the estimator (19) that it approximates; only $\hat{\Theta}_{SS}$ itself, computed from the closed-form $D^2\Theta$ of Theorem 3.1, is covered on the full range $s > (d+1)/2$. The gap may well be an artifact of the crude bound $\phi^{(4)} \lesssim \|\Delta\|_{C^1}^4$ assumed in Assumption 4.1, which charges a gradient factor $K_n^{1/d}$ to each of the four directions. The second derivative (10)–(11) charges only* one *gradient across its two directions. Substituting $\|\Delta\|_\infty \asymp \sqrt{K_n/n}$ for each direction that carries no gradient gives the ladder*

$$\begin{aligned}
\phi^{(4)} &\lesssim \|\Delta\|_{C^1}^4 && \text{(4 gradients)} \implies s > \tfrac{2(d+1)}{3},\\
\phi^{(4)} &\lesssim \|\Delta\|_\infty \|\Delta\|_{C^1}^3 && \text{(3 gradients)} \implies s > \tfrac{2d+1}{3},\\
\phi^{(4)} &\lesssim \|\Delta\|_\infty^2 \|\Delta\|_{C^1}^2 && \text{(2 gradients)} \implies s > \tfrac{2d}{3},
\end{aligned}$$

the last of which is weaker *than $s > (d+1)/2$ for every $d \geq 2$, so the gap would close entirely. Which of the three holds is decided by the third and fourth shape derivatives of Θ, which we have not computed; v1 asserted the $o_p(r_n^*)$ conclusion without either the assumption or the calculation. Until then the honest statement is: (21) is exact for quadratic paths, has a fourth-order error in general, and is provably rate-optimal under (22).*

Representation (21) is how we implement (19): it requires *no* numerical differentiation step size, no explicit computation of curvatures, normals, or corner angles, and it automatically includes *every* term of (10) — both margin diagonals, both diagonal corner terms, and the cross corner term — because the functional itself is evaluated along the joint direction. It is a half-sample generalized jackknife: the correction subtracts the "curvature" of Θ between the two independent half-sample fits.

Remark 4.4 (The jackknife identity is not a curvature correction for non-smooth first stages). *Lemma 4.2 is a statement about a smooth path. If $\hat\gamma^A, \hat\gamma^B$ are piecewise-constant (boosted trees, random forests), then $\phi(t) = \Theta(\bar\gamma + t\Delta)$ is a step function of t: it has no derivatives at all, $\phi^{(4)}$ does not exist, and Assumption 4.1 fails at every γ in the relevant neighborhood. Formula (21) remains a well-defined* estimator *— three evaluations of Θ — but it is no longer an approximation to $\frac{1}{8}D^2\Theta[\Delta,\Delta]$, and neither Theorem 4.5 nor Theorem 5.5 covers it. This is a prediction, not a caveat: the doubly debiased estimator of Section 5, which applies (21) to gradient-boosted fits, over-corrects and inflates dispersion in exactly the way a finite-difference formula does when the function differenced is not smooth (Table 2, DD rows).*

Perturbing one contrast at a time isolates the corner. Define the *margins-only* correction through the polarization decomposition

$$\begin{aligned} q_{\text{full}} &:= 4\big[\Theta(\hat\gamma^A) + \Theta(\hat\gamma^B) - 2\Theta(\bar\gamma)\big], \\ q_g &:= 4\big[\Theta(\hat g^A, \bar h) + \Theta(\hat g^B, \bar h) - 2\Theta(\bar\gamma)\big], \qquad q_h := 4\big[\Theta(\bar g, \hat h^A) + \Theta(\bar g, \hat h^B) - 2\Theta(\bar\gamma)\big], \\ \hat\Theta_{SS}^{\text{mar}} &:= \Theta(\bar\gamma) - \tfrac{1}{8}(q_g + q_h), \qquad \widehat{\text{corner}} := \tfrac{1}{8}\,(q_{\text{full}} - q_g - q_h). \end{aligned} \tag{23}$$

Perturbing g alone corresponds to the direction $(\Delta_g, 0)$, and $D^2\Theta[(\Delta_g,0),(\Delta_g,0)] = Q_g[\Delta_g,\Delta_g]$ *including* the $\cos\omega$ corner diagonal that (11) attaches to Q_g. Hence q_g and q_h absorb both corner diagonals, and by bilinearity

$$q_{\text{full}} - q_g - q_h \;\longrightarrow\; 2\,C[\Delta_g, \Delta_h], \qquad C[u,v] := \int_{\mathcal{C}} \frac{u\,v\,w}{\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|}\,\mathrm{d}\mathcal{H}^{d-2}, \tag{24}$$

the corner *cross* form alone. Consequently $\hat\Theta_{SS} = \hat\Theta_{SS}^{\text{mar}} - \widehat{\text{corner}}$, and $\widehat{\text{corner}}$ is a direct empirical measure of the corner cross term's stochastic diagonal — *not* of the whole of (17). Identity (24) is verified numerically to full precision on thirteen closed-form designs in Appendix C; it matters because it determines exactly which part of the corner survives margins-only debiasing (Proposition 4.7).

Theorem 4.5 (SS debiasing attains the minimax rate under weak smoothness). *Let Assumptions 1.4–1.5 and 4.1 hold, let the four arm regressions use sieve dimension $K_n \asymp n^{d/(2s+1)}$ (up to logs), and suppose $s > (d+1)/2$. Suppose in addition that the third-order remainder in (9) satisfies $R_3 = O_p(\|\bar\Delta\|_{C^1}^3) = o_p(r_n^*)$. Then*

$$\hat\Theta_{SS} - \Theta_0 = O_p(r_n^*), \qquad r_n^* = n^{-s/(2s+1)}, \qquad \frac{\sqrt{n}\,(\hat\Theta_{SS} - \Theta_0)}{\hat\sigma_n} \xrightarrow{d} \mathcal{N}(0,1),$$

with $\hat\sigma_n$ the two-band sieve variance (8) of the full-sample fit (the sieve variance remains *ratio-consistent for the debiased estimator). The same conclusion holds for the tuning-free implementation* $\hat{\Theta}_{SS}^{\text{jack}}$ *of* (21) *under the additional condition* (22)*; see Remark 4.3.*

Proof sketch. Expand $\Theta(\bar{\gamma})$ around γ_0 to second order and substitute (20). The linear term $D\Theta[\bar{\Delta}]$ is the average of two independent half-sample linear terms and delivers the CLT with the same σ_n. The quadratic term decomposes as

$$\tfrac{1}{2}D^2\Theta[\bar{\Delta},\bar{\Delta}] - \tfrac{1}{8}D^2\Theta[\Delta,\Delta] = \underbrace{\tfrac{1}{2}D^2\Theta[b,b]}_{\text{(i)}} + \underbrace{D^2\Theta[b,\bar{u}]}_{\text{(ii)}} + \underbrace{\tfrac{1}{2}D^2\Theta[u^A,u^B]}_{\text{(iii)}},$$

the stochastic diagonals having cancelled exactly. By Table 1: (i) is $O(K_n^{-(2s-1)/d}) = o(r_n^*)$ under $s > 1$; (ii) is $O_p(K_n^{(1-s)/d}\sqrt{K_n/n}) = O_p(r_n^*)$ iff $s \geq (d+1)/2$; and (iii) is a mean-zero degenerate U-statistic of order $n^{-1}K_n^{(d+3)/(2d)} = O_p(r_n^*)$ under the same condition (Appendix B). Thus *the binding constraint $s > (d+1)/2$ comes from the margin terms (ii) and (iii)*; the corner analogues of both are smaller by $K_n^{1/(2d)}$ or more and impose no binding requirement beyond transversality. The remaining pieces are R_3, assumed negligible, and — for the jackknife implementation — the fourth-order error of Lemma 4.2. Ratio consistency of $\hat{\sigma}_n$ follows from Proposition 2.3 because the correction changes the estimator by $o_p(\sigma_n/\sqrt{n})$. □

Remark 4.6 (What "$s > (d+1)/2$" is, and is not)**.** *v1 asserted that at the rate-optimal K_n the condition $s > (d+1)/2$ is "exactly" $K_n^{1/2-s/d} \to 0$. That is false: $K_n^{1/2-s/d} = n^{(d-2s)/(2(2s+1))} \to 0$ iff $s > d/2$. The correct equivalent statement in terms of the estimated geometry is*

$$\frac{K_n^{1/2+1/d}}{\sqrt{n}} \longrightarrow 0 \qquad \Longleftrightarrow \qquad s > \frac{d+1}{2} \quad \text{at } K_n \asymp n^{d/(2s+1)}, \tag{25}$$

i.e. uniform consistency of the gradient *of the estimated surface (each derivative costing $K_n^{1/d}$ on top of the $\sqrt{K_n/n}$ pointwise stochastic size). It is not a coincidence that the same threshold appears in the proof of Theorem 4.5: both terms (ii) and (iii) there carry exactly one gradient factor.*

Proposition 4.7 (Margins-only debiasing is insufficient)**.** *Under the conditions of Theorem 4.5 but with the margins-only correction* (23)*,*

$$\hat{\Theta}_{SS}^{mar} - \hat{\Theta}_{SS} = \widehat{corner},$$

and, by (24)*, the error of* $\hat{\Theta}^{mar}_{SS}$ *retains the corner* cross *term and nothing else of the corner:*

$$\mathcal{B}^{\times}_{\mathcal{C}} := \int_{\mathcal{C}} \frac{V_{gh}(x)\, w(x)}{\|\nabla g_0\|\, \|\nabla h_0\|\, |\sin \omega(x)|}\, \mathrm{d}\mathcal{H}^{d-2}(x) = \underbrace{\mathcal{B}^{\times,\mathrm{stoch}}_{\mathcal{C}}}_{\asymp |\overline{\rho\sigma_S\sigma_Y}|\, K_n/n} + \underbrace{\mathcal{B}^{\times,\mathrm{bias}}_{\mathcal{C}}}_{\asymp K_n^{-2s/d} = r_n^{*2}} .$$

The two $\cos\omega$ *corner diagonals of* (11) *are components of* Q_g *and* Q_h *and* are *removed by the margins-only correction. The split-sample statistic* $\widehat{\mathrm{corner}}$ *is unbiased for* $\mathcal{B}^{\times,\mathrm{stoch}}_{\mathcal{C}}$ *only: because the common approximation bias cancels from* $\Delta = \hat{\gamma}^A - \hat{\gamma}^B$*,* $\mathbb{E}[\widehat{\mathrm{corner}}]$ *contains no bias-product term. The bias part* $\mathcal{B}^{\times,\mathrm{bias}}_{\mathcal{C}}$ *is* $o(r_n^*)$ *and is removed by undersmoothing, not by debiasing. Consequently, provided* $\overline{\rho\sigma_S\sigma_Y} \not\equiv 0$ *on* $\mathcal{C}$*,* $\hat{\Theta}^{mar}_{SS}$ *carries a residual bias* $\asymp K_n/n$*, which* is not $o(r_n^*)$ *when* $s \le d-1$*; since the SS guarantee of Theorem 4.5 extends down to* $s > (d+1)/2$*, margins-only debiasing forfeits the weak-smoothness guarantee on the window* (16)*, nonempty whenever* $d \ge 4$*. If the two contrast errors are conditionally uncorrelated at every corner, margins-only and corner-aware debiasing agree to this order. For* $d \in \{2,3\}$ *the window is empty and the corner cross term is a finite-sample constants effect — of order at most* $K_n^{-1/d}$ *times the margin diagonals' bound, and by Remark 3.3 plausibly of the* same *order as the margin diagonals actually are on a generic geometry.*

Proof. The first display is the definition of $\widehat{\mathrm{corner}}$ in (23). For the second, apply the decomposition in the proof of Theorem 4.5 to $D^2\Theta[\bar{\Delta}, \bar{\Delta}] - \frac{1}{4} D^2\Theta[(\Delta_g, 0), (\Delta_g, 0)] - \frac{1}{4} D^2\Theta[(0, \Delta_h), (0, \Delta_h)]$: by bilinearity the Q_g and Q_h stochastic diagonals cancel in full — including their corner components — leaving $C[\bar{\Delta}_g, \bar{\Delta}_h]$, whose expectation is $\mathcal{B}^{\times}_{\mathcal{C}}$. Splitting $\bar{\Delta} = b + \bar{u}$ and using $\mathbb{E}[\bar{u}] = 0$ separates it into the stated stochastic and bias parts, of orders $|\overline{\rho\sigma_S\sigma_Y}|\, K_n/n$ and $K_n^{-2s/d}$ by (14). That $\mathbb{E}[\widehat{\mathrm{corner}}]$ omits the bias part is immediate from $\Delta = u^A - u^B$ in (20). □

Remark 4.8 (Relation to v1)**.** *v1's Proposition on margins-only debiasing stated* $\mathbb{E}[\widehat{corner}] \asymp 2C[b_g, b_h] + \rho_{SY} K_n/n$*. The identification of the leftover with the* ρ_{SY}*-driven* cross *term is correct — an earlier revision of the present draft wrongly reassigned the* $\cos\omega$ *diagonals to it, and* (24) *settles the matter. What does not belong is the bias-product term in* $\mathbb{E}[\widehat{corner}]$*, whose presence contradicted v1's own (correct) description of Table 4. Proposition 4.7 keeps the two objects separate: the* residual bias *of* $\hat{\Theta}^{mar}_{SS}$ *contains both parts; the* SS estimate of it *targets only the stochastic part. The empirical "remainder" column of Table 4 is precisely* $\mathcal{B}^{\times,\mathrm{bias}}_{\mathcal{C}}$*, and its near-constancy in* n *at fixed* K *is the corresponding prediction.*

The pair (Theorem 4.5, Proposition 4.7) is the central methodological message of the two-threshold extension: *quadratic debiasing must be corner-aware.* Because the jackknife form (21) perturbs both contrasts jointly, it is corner-aware by construction — the theory identifies what would go wrong with the seemingly natural surface-by-surface application of the single-threshold correction.

Remark 4.9 (Leave-one-out analogue)**.** *The LOO estimator replaces* $\frac{1}{8}D^2\Theta[\Delta,\Delta]$ *by*

$$\frac{1}{2n^2}\sum_{i=1}^{n} D^2\Theta(\hat\gamma)\big[(s_i\hat\epsilon_{S,i}^{(-i)},\, s_i\hat\epsilon_{Y,i}^{(-i)}),\ (s_i\hat\epsilon_{S,i}^{(-i)},\, s_i\hat\epsilon_{Y,i}^{(-i)})\big],$$

whose corner cross component carries the product $\hat\epsilon_{S,i}^{(-i)}\hat\epsilon_{Y,i}^{(-i)}$ *— an explicit estimate of the* ρ_{SY}*-weighted part of* (17) *— while its corner diagonal components carry* $(\hat\epsilon_{S,i}^{(-i)})^2$ *and* $(\hat\epsilon_{Y,i}^{(-i)})^2$*, which is why the LOO correction, like the SS correction, captures the* $\cos\omega$ *terms as well. It attains the same guarantees as Theorem 4.5 under the corresponding LOO conditions of Chen and Gao (2026); we focus on SS in the simulations because* (21) *is tuning-free.*

5 Sieve DML and the Doubly Debiased Estimator

5.1 Cross-fitted two-band sieve-Riesz correction

The debiasing of Section 4 targets the *second-order* remainder but retains sieve LS first stages. The complementary device — the sieve-Riesz DML of the extended single-threshold draft — targets the *first-order* term on the sieve space and thereby accommodates generic machine-learning first stages. Let $(I_\kappa)_{\kappa=1}^K$ partition the sample, $\hat\mu_{-\kappa}$ be *any* first-stage learner of the four arm means fit off fold κ, and $\hat v^*_{K_n,o,-\kappa}$ the feasible two-band sieve-Riesz representer (6)–(7) computed *off fold* κ — both its Gram matrix and its bands must use only $I_{-\kappa}$. The cross-fitted sieve-debiased estimator is

$$\tilde\Theta := \frac{1}{K}\sum_{\kappa=1}^{K}\left[\Theta(\hat\gamma_{-\kappa}) + \frac{K}{n}\sum_{i\in I_\kappa}\sum_{o\in\{S,Y\}} \hat v^*_{K_n,o,-\kappa}(X_i,W_i)\big(O_i - \hat\mu_{o,-\kappa}(X_i,W_i)\big)\right], \tag{26}$$

with $O_i \in \{S_i, Y_i\}$. The correction annihilates the first-order estimation error of $\hat\mu_{-\kappa}$ *on the sieve space*: it is the local influence function of the irregular functional, well defined even though no global influence function exists.

Assumption 5.1 (Two-band local robustness)**.** *For each fold* κ*: (a)* $\|\hat\mu_{-\kappa}-\mu_0\|_\infty = o_p(1)$ *and* $\|\hat\gamma_{-\kappa}-\gamma_0\|_\infty = o_p(\varepsilon_n)$*; (b) the band sequence satisfies* $\varepsilon_n \downarrow 0$*,* $n\varepsilon_n/K_n \to \infty$*, and* $\varepsilon_n^{-1}\|\hat\gamma_{-\kappa}-\gamma_0\|_\infty = o_p(1)$*; (c) (*boundary stability*) the estimated level sets* $\{\hat g_{-\kappa}=0\}$*,* $\{\hat h_{-\kappa}=0\}$ *satisfy Assumption 1.5(a)–(b) with constants bounded away from zero w.p.* $\to 1$*, and* $\mathcal{H}^{d-1}(\{\hat g_{-\kappa}=0\}\triangle\{g_0=0\}) = o_p(1)$*; (d) the product rate* $\left\|\hat v^*_{K_n,-\kappa} - v^*_{K_n}\right\|\cdot\|\hat\mu_{-\kappa}-\mu_0\| =$

$o_p(\sqrt{K_n^{1/d}/n})$; *(e) the sieve approximation bias of the representer is* $o(\sqrt{K_n^{1/d}/n})$, *and the out-of-span boundary remainder* R_5 *of* (27) *is* $o_p(\sqrt{K_n^{1/d}/n})$; *(f) the second-order remainder of* $\Theta(\hat\gamma_{-\kappa})$ *is* $o_p(\sqrt{K_n^{1/d}/n})$, *i.e.* $\|\hat\gamma_{-\kappa}-\gamma_0\|_{C^1}^2 \cdot \|\hat\gamma_{-\kappa}-\gamma_0\|_\infty = o_p(\sqrt{K_n^{1/d}/n})$; *(g)* $n^{-\varsigma/2}\,\mathbb{E}[\left|v^*_{K_n}(X,W)\epsilon\right|^{2+\varsigma}]/\sigma_n^{2+\varsigma} \to 0$ *for some* $\varsigma > 0$ *(Lyapunov, for the triangular array* $n^{-1/2}\sigma_n^{-1}\sum_i v^*_{K_n}\epsilon_i$*) and* $\liminf \sigma_n^2/K_n^{1/d} > 0$ *(nondegeneracy of the stacked variance).*

Theorem 5.2 (Two-band sieve DML)**.** *Under Assumptions 1.4–1.5 and 5.1,*

$$\frac{\sqrt{n}\,(\tilde\Theta-\Theta_0)}{\sigma_n} \xrightarrow{d} \mathcal{N}(0,1), \qquad \sigma_n^2 \asymp K_n^{1/d},$$

and the two-band variance estimator (8), *computed with cross-fitted residuals, is ratio-consistent. As in the single-threshold case, the irregularity* weakens *the product-rate requirement relative to regular DML: the tolerance is* $\sqrt{K_n^{1/d}/n} \gg n^{-1/2}$.

Assumption 5.1 is longer than the corresponding list in v1, which cited only sup-norm consistency, a product rate, and sieve approximation bias. Those three do not control a boundary functional: (b) and (c) are what make the ε-band a consistent surrogate for a Hausdorff integral at an *estimated* surface, (f) is the nonlinear (moving-boundary) remainder that has no analogue in regular DML, and (g) is what a CLT for a triangular array with $\sigma_n^2 \to \infty$ requires. We state them because the two conditions that bind in our experiments are (e) and (f), not the product rate.

Remark 5.3 (Least-squares orthogonality; why raw AIPW corrections miscalibrate)**.** *Suppose the first stage* is *the sieve LS estimator on the same basis used for the representer, and consider the* non-cross-fitted *estimator: a full-sample fit with in-sample residuals. Then the normal equations* $\sum_i \psi(X_i,W_i)\hat\epsilon_i = 0$ *hold exactly, and since* $\hat v^*$ *lies in the span of* ψ *the correction vanishes identically: the sieve plug-in is automatically sieve-debiased at first order. This explains a phenomenon documented in our earlier experiments: adding a* raw *ε-band AIPW correction* $(2\varepsilon)^{-1}\mathbb{1}\{|\hat\tau| < \varepsilon\}\xi_i$ *on top of a full-sample sieve plug-in over-corrects — the projected (sieve-Riesz) correction it approximates is exactly zero, and the discrepancy is pure noise with a* $1/\varepsilon$ *variance inflation.*

This exactness does not survive cross-fitting*, contrary to what v1 asserted. In* (26) *the residuals* $O_i - \hat\mu_{-\kappa}(X_i)$ *for* $i \in I_\kappa$ *are* out-of-fold*: the normal equations hold on* $I_{-\kappa}$*, not on* I_κ*, so the correction is a nondegenerate mean-zero term of the same order* $\sqrt{K_n^{1/d}/n}$ *as the leading variance. With a sieve LS first stage the cross-fitted correction is therefore not zero but merely* redundant*: it substitutes one* $O_p(\sqrt{K_n^{1/d}/n})$ *term for another without changing the limit distribution. The projected correction* (26) *is* useful *precisely when the first stage is* *not the LS fit on the representer's basis — e.g. gradient boosting or random forests — which is its intended use case.*

Remark 5.4 (When the sieve cannot see the learner's error)**.** *Assumption 5.1(e) deserves emphasis because it is the one that binds in practice. Write* $\Delta := \hat{\mu}_{-\kappa} - \mu_0$ *and decompose the plug-in's first-order error as*

$$D_\mu \Theta(\mu_0)[\Delta] = \underbrace{\left\langle v^*_{K_n}, \Delta \right\rangle}_{\textit{in-span}} + \underbrace{D_\mu \Theta(\mu_0)\left[(I - \Pi_{K_n})\Delta\right]}_{=:R_5,\ \textit{out-of-span}}. \tag{27}$$

The correction in (26) replaces the first term by the score $\mathbb{E}_n[v^*\epsilon]$*;* R_5 *is untouched. If* $\hat{v}^*_{-\kappa}$ *and* $\hat{\mu}_{-\kappa}$ *are measurable with respect to* $I_{-\kappa}$ *and the score is conditionally mean zero, then* $\mathrm{Cov}(\textit{score}, R_5) = 0$ *exactly, so*

$$\mathrm{Var}(\tilde{\Theta}) = \frac{\sigma_n^2}{n} + \mathrm{Var}(R_5) + \mathrm{Var}(\textit{h.o.t.}) + \textit{cross terms in the h.o.t.},$$

while $\hat{\sigma}_n^2$ *estimates only* σ_n^2*.* This is a statement about which terms the standard error omits; it is not a statement about the sign of the coverage error*, and v1's conclusion that the interval "can only under-cover, never over-cover" does not follow, for three reasons. First, the plug-in error* also *contains* R_5*; the net change in dispersion from adding the correction is* $\sigma_n^2/n - \mathrm{Var}(\langle v^*_{K_n}, \Delta \rangle)$*, of either sign, so the debiased estimator is not "the plug-in plus an independent perturbation" except in the degenerate case* $\Pi_{K_n}\Delta \equiv 0$*. Second, coverage depends on bias as well as dispersion, and* $\hat{\sigma}_n$ *is itself a noisy band-based estimate whose own bias — upward at small* ε_n*, downward when the band is widened — is not sign-restricted; SE/SD ratios above one occur in several cells of Table 2. Third, the exact-orthogonality step requires the representer to be built off fold* κ*, which our current implementation does not do (Section 6.4, item (L6)).*

Directly measured, the variance-addition channel is small on our design: the saved variance experiment reports $\{\mathrm{Var(DML)} - \mathrm{Var(plug\text{-}in)}\}/\widehat{\mathrm{SE}}^2$ *equal to* -0.04 *(GBR),* -0.00 *(RF) and* $+0.04$ *(smooth RBF ridge) at* $n = 4000$ *— not the* $\approx +1$ *that the "independent added variance" reading predicts.* The machine-learning undercoverage on this design is therefore predominantly residual bias, not unmodelled variance. *We correct v1, which asserted the opposite.*

What must be small in Assumption 5.1(e) is not $\|(I - \Pi_{K_n})\Delta\|_{L^2}$ *but the* boundary *functional evaluated at it: only the sieve's ability to track the learner's error* on the margins *matters, and global* L^2 *richness is not the right diagnostic. This is exactly where regression trees fail. A boosted ensemble's error is piecewise constant with jumps at adaptively chosen* *split hyperplanes, and boosting places its splits where the loss gradient is largest — for a CATE surface whose zero level set defines the estimand, that is near the margins themselves. So* $(I - \Pi_{K_n})\Delta$ *is largest exactly on the set the derivative integrates over. Section 6 measures a global-*L^2 *proxy for this: the* R^2 *of the boosted first-stage error on the tensor B-spline representer span is* 0.21–0.34 *across the four arm regressions (mean* 0.26*), and the projected correction removes about* 31% *of the plug-in's boundary bias. Tree ensembles additionally violate the smoothness conditions in a way that is not merely quantitative:* $\nabla(\hat{\mu} - \mu_0)$ *is a sum of Dirac measures on the split hyperplanes, the* ε*-band* $\{|\hat{\tau}| < \varepsilon\}$ *is a union of whole leaf cells (of measure* $O(1)$ *or exactly zero, never* $O(\varepsilon)$*), and the estimated region is an axis-aligned staircase whose surface measure over-counts an oblique boundary by up to* $\sqrt{d}$*. The practical conclusion is stated in Section 6.3: match the representer basis to the learner's own feature map, and prefer smooth learners.*

The diagnosis is not special to two thresholds. In the single-threshold setting, the extended draft's own Table 7 reports, for the value functional on identical samples and folds, a gradient-boosting DML estimator that is essentially *unbiased* ($|\text{bias}| \leq 0.003$ at every n) yet covers only 0.90, with SE/SD $= 0.78, 0.84, 0.88$ — while a smooth random-feature first stage on the same design has SE/SD $= 0.84, 0.94, 0.99$ and coverage rising to 0.93. The decomposition here explains that contrast, though — given the variance measurement above — the mechanism in the two-threshold design is the omitted-bias channel rather than the omitted-variance channel.

5.2 The doubly debiased estimator

The two devices remove different terms — the first-order projection error (DML) and the second-order diagonal (SS) — and can be composed. Split the sample into halves A, B; fit generic ML first stages on each half; and let $\hat{\gamma}^A, \hat{\gamma}^B$ be the two ML contrast pairs (each predicted on the full covariate sample), $\bar{\gamma}$ their average. The *doubly debiased* (DD) estimator is

$$\hat{\Theta}_{DD} := \underbrace{2\,\Theta(\bar{\gamma}) - \tfrac{1}{2}\big(\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B)\big)}_{\text{SS quadratic correction (21) on ML fits}} + \underbrace{\frac{1}{n}\sum_{i=1}^{n}\sum_{o\in\{S,Y\}} \hat{v}^*_{K_n,o,-\kappa(i)}(X_i, W_i)\,\hat{\epsilon}^{\text{oof}}_{o,i}}_{\text{cross-fitted two-band Riesz correction}}, \tag{28}$$

where $\hat{\epsilon}^{\text{oof}}_{o,i}$ is the out-of-fold residual of observation i (predicted by the half not containing i), $\kappa(i)$ is i's fold, and $\hat{v}^*_{\cdot,-\kappa(i)}$ is built off that fold on a linear Riesz basis (tensor B-splines for moderate d, random features for large d) with bands evaluated at the cross-fitted contrasts. Both corrections are computed from the same two half-sample fits, so DD costs one cross-fit beyond the SS estimator.

Theorem 5.5 (Doubly debiased estimation)**.** *Suppose Assumption 5.1 holds for the ML first stages with condition (f)* deleted*, and in addition: (i) Assumption 4.1 holds along the segment* $\bar{\gamma} + t\Delta$, $|t| \le \frac{1}{2}$, *w.p.* $\to 1$*; (ii)* $\|\hat{\gamma}^j - \gamma_0\|_{C^1} = o_p(1)$ *and the two half-sample fits are independent; (iii) the third-order margin/corner remainders and the fourth-order jackknife error of Lemma 4.2 are* $o_p(\sqrt{K_n^{1/d}/n})$*. Then*

$$\frac{\sqrt{n}\,(\hat{\Theta}_{DD} - \Theta_0)}{\hat{\sigma}_n} \xrightarrow{d} \mathcal{N}(0,1),$$

with $\hat{\sigma}_n$ *from (8) on cross-fitted residuals. The point of deleting Assumption 5.1(f) and replacing it by (iii) is exactly the gain: the quadratic correction removes the own-half second-order diagonal of the ML errors — margins* and *corner — so the smallness requirement on the learners drops from second order to third order. Relative to Theorem 4.5, generic learners replace the sieve LS first stage. Note that the familiar regular-DML rate* $\|\hat{\gamma}^j - \gamma_0\|_\infty = o_p(n^{-1/4})$ *is* not *imposed: irregularity already relaxes the second-order tolerance from* $n^{-1/2}$ *to* $\sqrt{K_n^{1/d}/n}$*, and the quadratic correction relaxes it again.*

Theorem 5.5 answers, in the two-threshold setting, the question left open in the single-threshold draft (its Remark 7): the cross-fitted sieve influence-function correction and the diagonal quadratic correction are *compatible*, because they operate on orthogonal terms of the same expansion — one linear (projected), one quadratic (diagonal). Condition (i) is the binding one and is not innocuous: by Remark 4.4 it *fails* for tree-ensemble first stages, so Theorem 5.5 does not cover the GBR-based DD estimator evaluated in Section 6. The practical value of each component is an empirical question: for smooth truths and sieve first stages both corrections are small (the linear one redundant by Remark 5.3); for rough truths the quadratic correction binds; for flexible-but-biased *smooth* ML first stages both bind.

6 Monte Carlo Evidence Calibrated to the Graduation Experiment

6.1 Design

Following the calibration philosophy of Chen and Ritzwoller (2023) — whose generative model is itself fit to the Banerjee et al. (2015) "graduation" RCT in Pakistan ($n = 854$; 2-year outcomes as surrogates S, 3-year consumption as Y) — we evaluate all estimators on three data-generating processes with *known* truth:

KRR a kernel-ridge exact-truth oracle: per-arm conditional means of (S, Y) fit to the graduation data by kernel ridge regression are taken as the *true* $\mu_{o,0}(\cdot, w)$; covariates are drawn from a Gaussian-KDE fit of the empirical covariate distribution ($d = 2$), truncated to $[-3, 3]^2$; residual noise is calibrated to the data with within-unit (S, Y) coupling ($\rho_{SY} \neq 0$). The truth $\theta_0 = 0.161$ and all margin/corner geometry are computable exactly. (See Section 6.4, items (L1)–(L3), for two implementation defects that displace this "exact" truth.)

WGAN the Chen and Ritzwoller (2023) three-GAN cascade $X \to S \mid X, W \to Y \mid S, X, W$ (the `ds-wgan` design of Athey et al., 2021) trained on the same data; the fitted generators are frozen as the truth and CATEs computed by common-random-number Monte Carlo. ReLU generators make the true contrast surfaces piecewise linear — effectively $s \approx 1$ — so this design sits *below* the smoothness threshold of every theorem in this paper and is a stress test, not a corroboration. We use two variants: the *scalar-surrogate* cascade, whose geometry has both margins binding ($\theta_0 = 0.194$), and the *full-surrogate* cascade at $d = 2$, which is *degenerate* — conditioning Y on the full 21-dimensional surrogate vector flattens τ_S and pushes $\tau_Y > 0$ everywhere, so $\theta_0 = 0$ with empty margins, violating Assumption 1.5(a) and (c). The degenerate arm is retained deliberately: it shows how the boundary-band inference behaves when the estimand sits on the boundary of the parameter space and no margin exists.

Affine a Gaussian–affine design with closed-form orthant truth, as a specification check of the numerical pipeline.

For each DGP we draw experiments of size $n \in \{1000, 2000, 4000, 8000\}$ (500 replications each) and compare, at nominal 95%: (i) the sieve plug-in $\hat{\Theta}$; (ii) margins-only SS $\hat{\Theta}^{\text{mar}}_{SS}$; (iii) corner-aware SS $\hat{\Theta}_{SS}$ (21); (iv) the 2-fold cross-fitted GBR plug-in; (v) (iv) + projected Riesz correction; (vi) the doubly debiased $\hat{\Theta}_{DD}$ (28). A separate decomposition experiment (Table 3) then varies the DML construction one axis at a time — fold count, first-stage learner (boosting, random forest, smooth random-feature ridge), and representer basis — to isolate the source of the machine-learning undercoverage. All are studentized by the two-band sieve variance (8).[1] Table 2 reports bias, MC standard deviation, RMSE, the mean-SE/MC-SD

[1] Implementation details: the sieve dimension is held *fixed* across n at $5 \times 5 = 25$ tensor B-spline basis functions per arm (two knot segments per coordinate), so the design varies n at fixed K rather than along $K_n \asymp n^{d/(2s+1)}$; see Section 6.4, item (L4). Bandwidths are $\varepsilon = \delta \cdot \widehat{\text{sd}}(\hat{\tau})$ with a trimmed standard deviation and $\delta = 0.08$ (sieve family) or 0.10 (ML family); if fewer than five evaluation points fall in a band it is widened by factors of 1.5 (at most six times) — an adaptivity that is active mainly on the degenerate arm, where the margins are empty. Ties are broken as $\mathbb{1}\{\hat{\tau}_S \geq 0\}$ and harm as the strict $\hat{\tau}_Y < 0$; the difference is measure-zero under Assumption 1.5.

ratio, empirical coverage, and mean CI length; the estimated corner correction is examined separately in Section 6.2. With 500 replications the Monte Carlo standard error of a coverage figure near 0.95 is about 0.010, and near 0.90 about 0.013; differences smaller than roughly 0.03 should not be read as real.

Table 2: Calibrated Monte Carlo (500 replications per cell): bias, MC standard deviation, RMSE, mean-SE/MC-SD ratio, empirical 95% coverage, and mean CI length, by DGP, estimator, and sample size. All intervals are studentized by the two-band sieve-Riesz standard error (8) (cross-fitted residuals for the GBR family). Sieve dimension is fixed at $K = 25$ per arm for every n.

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
KRR (smooth truth), $\theta_0 = 0.161$							
Plug-in (sieve)	1000	+0.0103	0.0362	0.0376	1.11	0.962	0.158
	2000	+0.0052	0.0259	0.0264	1.06	0.958	0.108
	4000	+0.0037	0.0185	0.0188	1.00	0.954	0.073
	8000	+0.0031	0.0133	0.0136	0.94	0.948	0.049
SS margins-only	1000	-0.0058	0.0456	0.0459	0.88	0.882	0.158
	2000	-0.0050	0.0294	0.0298	0.94	0.918	0.108
	4000	-0.0021	0.0204	0.0205	0.91	0.932	0.073
	8000	-0.0005	0.0135	0.0135	0.93	0.926	0.049
SS corner-aware	1000	+0.0030	0.0446	0.0447	0.90	0.894	0.158
	2000	-0.0005	0.0293	0.0293	0.94	0.932	0.108
	4000	+0.0004	0.0204	0.0204	0.91	0.934	0.073
	8000	+0.0007	0.0134	0.0134	0.93	0.930	0.049
Cross-fit GBR	1000	+0.0666	0.0316	0.0737	1.41	0.680	0.175
	2000	+0.0444	0.0238	0.0504	1.06	0.578	0.099
	4000	+0.0239	0.0189	0.0304	0.87	0.666	0.064
	8000	+0.0088	0.0142	0.0167	0.82	0.820	0.046
+ Riesz corr.	1000	+0.0562	0.0545	0.0783	0.82	0.704	0.175
	2000	+0.0260	0.0268	0.0373	0.94	0.784	0.099
	4000	+0.0145	0.0196	0.0244	0.83	0.808	0.064
	8000	+0.0045	0.0141	0.0148	0.83	0.882	0.046
+ quad. (DD)	1000	+0.0488	0.0653	0.0815	0.68	0.706	0.175
	2000	+0.0061	0.0361	0.0366	0.70	0.830	0.099
	4000	-0.0046	0.0273	0.0277	0.60	0.752	0.064

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
	8000	-0.0109	0.0195	0.0223	0.60	0.690	0.046
WGAN scalar (rough truth), $\theta_0 = 0.194$							
Plug-in (sieve)	1000	+0.0645	0.1008	0.1195	1.34	0.956	0.527
	2000	+0.0592	0.0884	0.1063	1.15	0.928	0.398
	4000	+0.0470	0.0618	0.0776	1.23	0.936	0.298
	8000	+0.0317	0.0453	0.0552	1.13	0.952	0.200
SS margins-only	1000	+0.0435	0.1396	0.1461	0.96	0.886	0.527
	2000	+0.0373	0.1236	0.1290	0.82	0.866	0.398
	4000	+0.0275	0.0782	0.0828	0.97	0.918	0.298
	8000	+0.0152	0.0537	0.0557	0.95	0.940	0.200
SS corner-aware	1000	+0.0655	0.1370	0.1518	0.98	0.876	0.527
	2000	+0.0519	0.1213	0.1318	0.84	0.868	0.398
	4000	+0.0363	0.0780	0.0859	0.98	0.910	0.298
	8000	+0.0206	0.0535	0.0573	0.96	0.930	0.200
Cross-fit GBR	1000	+0.0452	0.0434	0.0626	2.17	0.960	0.369
	2000	+0.0463	0.0359	0.0585	1.18	0.840	0.166
	4000	+0.0469	0.0318	0.0567	1.03	0.726	0.128
	8000	+0.0415	0.0276	0.0499	0.97	0.656	0.105
+ Riesz corr.	1000	+0.0492	0.1242	0.1335	0.76	0.910	0.369
	2000	+0.0411	0.0417	0.0585	1.02	0.840	0.166
	4000	+0.0456	0.0335	0.0566	0.98	0.718	0.128
	8000	+0.0416	0.0280	0.0501	0.95	0.632	0.105
+ quad. (DD)	1000	+0.0609	0.1413	0.1537	0.67	0.772	0.369
	2000	+0.0446	0.0703	0.0832	0.60	0.656	0.166
	4000	+0.0336	0.0570	0.0661	0.57	0.660	0.128
	8000	+0.0209	0.0446	0.0492	0.60	0.702	0.105
WGAN full-surrogate (degenerate), $\theta_0 = 0.000$							
Plug-in (sieve)	1000	+0.0604	0.0432	0.0742	1.60	0.922	0.271
	2000	+0.0329	0.0243	0.0409	1.52	0.940	0.144
	4000	+0.0173	0.0113	0.0207	1.73	0.936	0.076
	8000	+0.0085	0.0061	0.0105	1.73	0.930	0.042
SS margins-only	1000	+0.0140	0.0669	0.0683	1.03	0.866	0.271
	2000	+0.0019	0.0362	0.0362	1.02	0.894	0.144
	4000	+0.0011	0.0188	0.0188	1.04	0.886	0.076
	8000	-0.0002	0.0092	0.0092	1.15	0.874	0.042
SS corner-aware	1000	+0.0314	0.0643	0.0715	1.07	0.888	0.271

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
	2000	+0.0101	0.0347	0.0361	1.06	0.906	0.144
	4000	+0.0050	0.0183	0.0190	1.06	0.898	0.076
	8000	+0.0018	0.0091	0.0093	1.16	0.898	0.042
Cross-fit GBR	1000	+0.1742	0.0357	0.1778	1.94	0.126	0.271
	2000	+0.1608	0.0300	0.1636	1.17	0.008	0.138
	4000	+0.1341	0.0231	0.1360	1.02	0.000	0.092
	8000	+0.0962	0.0175	0.0977	0.92	0.000	0.063
+ Riesz corr.	1000	+0.1732	0.1099	0.2051	0.63	0.146	0.271
	2000	+0.1598	0.0361	0.1638	0.97	0.026	0.138
	4000	+0.1342	0.0241	0.1363	0.98	0.000	0.092
	8000	+0.0966	0.0176	0.0982	0.91	0.000	0.063
+ quad. (DD)	1000	+0.1184	0.1203	0.1687	0.57	0.422	0.271
	2000	+0.0858	0.0523	0.1005	0.67	0.376	0.138
	4000	+0.0461	0.0353	0.0580	0.67	0.508	0.092
	8000	+0.0071	0.0200	0.0212	0.80	0.866	0.063
Affine (exact truth), $\theta_0 = 0.209$							
Plug-in (sieve)	1000	+0.0042	0.0546	0.0547	0.97	0.916	0.208
	2000	+0.0017	0.0339	0.0339	1.07	0.958	0.142
	4000	+0.0012	0.0249	0.0249	1.00	0.940	0.098
	8000	-0.0013	0.0171	0.0172	1.00	0.954	0.067
SS margins-only	1000	+0.0023	0.0680	0.0680	0.78	0.858	0.208
	2000	+0.0007	0.0392	0.0391	0.93	0.934	0.142
	4000	+0.0003	0.0267	0.0267	0.93	0.934	0.098
	8000	-0.0022	0.0178	0.0179	0.96	0.930	0.067
SS corner-aware	1000	+0.0025	0.0681	0.0680	0.78	0.860	0.208
	2000	+0.0005	0.0393	0.0393	0.92	0.928	0.142
	4000	+0.0003	0.0268	0.0268	0.93	0.932	0.098
	8000	-0.0022	0.0178	0.0179	0.96	0.932	0.067
Cross-fit GBR	1000	+0.0277	0.0378	0.0468	1.00	0.862	0.148
	2000	+0.0161	0.0284	0.0326	0.94	0.878	0.105
	4000	+0.0097	0.0238	0.0257	0.85	0.886	0.079
	8000	+0.0036	0.0174	0.0178	0.86	0.900	0.059
+ Riesz corr.	1000	+0.0255	0.0416	0.0487	0.91	0.870	0.148
	2000	+0.0140	0.0291	0.0322	0.92	0.896	0.105
	4000	+0.0077	0.0241	0.0253	0.84	0.876	0.079
	8000	+0.0022	0.0182	0.0184	0.83	0.886	0.059

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
+ quad. (DD)	1000	+0.0032	0.0604	0.0604	0.63	0.764	0.148
	2000	-0.0024	0.0441	0.0441	0.60	0.778	0.105
	4000	-0.0014	0.0325	0.0325	0.62	0.780	0.079
	8000	-0.0036	0.0241	0.0243	0.63	0.778	0.059

Figure 1: Empirical 95% coverage by sample size, DGP (columns), and first-stage family (rows). Dashed line: nominal 0.95.

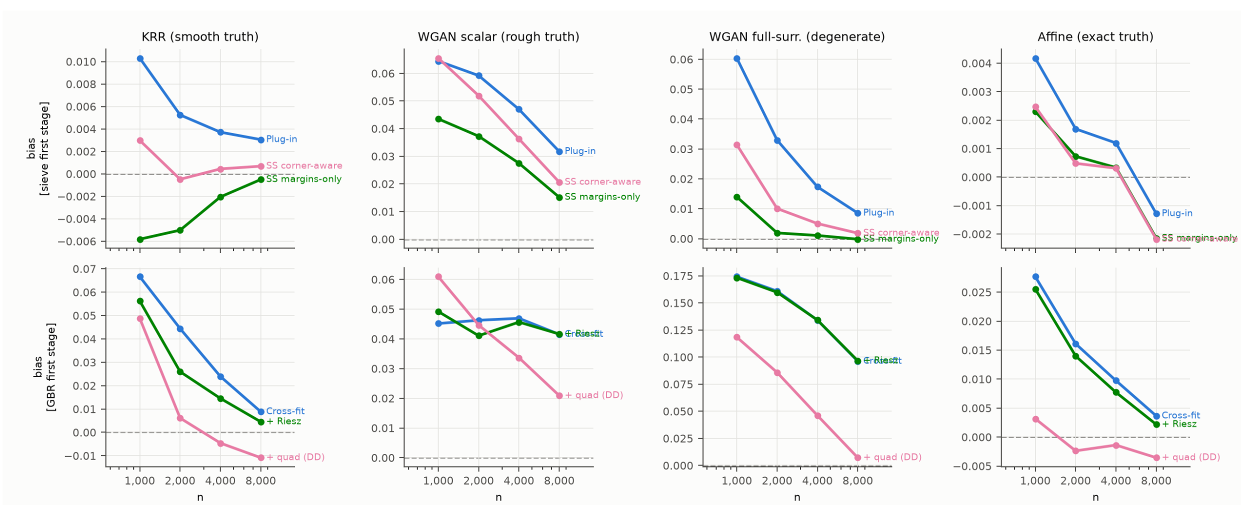

Figure 2: Monte Carlo bias by sample size, DGP, and estimator. Dashed line: zero.

Table 3: Decomposition of the machine-learning undercoverage on the KRR design ($n = 4000$, 300 replications). "Plug-in bias" is the uncorrected cross-fit bias; "bias", "SD", "SE/SD", and "coverage" refer to the Riesz-corrected estimator, studentized by (8). The tensor-sieve plug-in is the reference. Varying folds, learner, and representer basis one at a time isolates each mechanism of Remark 5.4. MC standard error of a coverage figure is about 0.013 at 300 replications.

Configuration	Bias	Plug-in bias	SD	SE/SD	Coverage
tensor-sieve plug-in	+0.0034	+0.0034	0.0178	1.03	0.943
gbr K=2 riesz=sieve	+0.0216	+0.0317	0.0186	0.96	0.760
gbr K=5 riesz=sieve	+0.0106	+0.0160	0.0191	0.94	0.903
gbr K=10 riesz=sieve	+0.0079	+0.0127	0.0196	0.92	0.913
rf K=5 riesz=sieve	+0.0119	+0.0453	0.0195	1.01	0.907
krr K=5 riesz=sieve	+0.0038	+0.0039	0.0184	0.90	0.917
gbr K=5 riesz=rf	+0.0078	+0.0160	0.0203	0.96	0.937
krr K=5 riesz=rf	+0.0047	+0.0039	0.0181	0.98	0.937
gbr(high-cap) K=5 riesz=sieve	+0.0439	+0.0461	0.0131	1.00	0.077
gbr K=5 no correction	+0.0160	+0.0160	0.0189	0.94	0.857

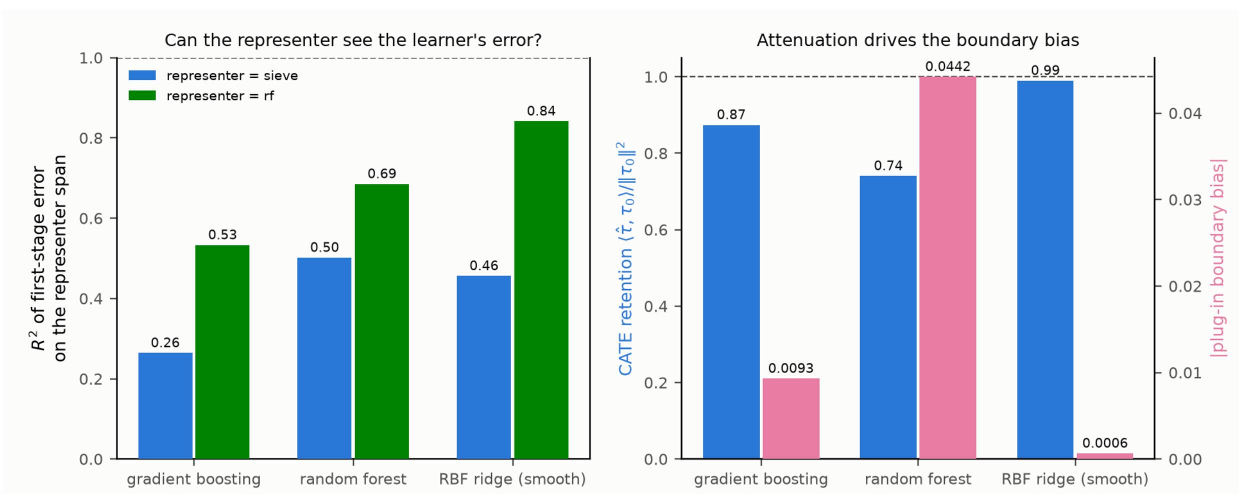

Figure 3: Why boosting-based DML undercovers. *Left*: the fraction (R^2) of the first-stage error that the sieve-Riesz representer can span in global L^2, measured against the oracle arm means; a boosted-tree error is largely invisible to a smooth representer, a random-feature ridge's error is not. *Right*: tree learners attenuate the CATE (no-intercept projection coefficient of fitted on true, $\hat{\tau}'\tau/\tau'\tau < 1$), which displaces the decision boundary and biases the plug-in; the smooth learner barely attenuates and is nearly unbiased. Both panels average five simulated datasets and are indicative rather than precise.

6.2 Corner anatomy

Because the KRR oracle's contrast surfaces are known in closed form, the corner set and its geometry are computable: the zero curves of τ_S and τ_Y cross at ten points, with gradient norms $\|\nabla\tau_S\| \in [10.3, 135.8]$, $\|\nabla\tau_Y\| \in [13.9, 56.4]$, $|\cos\omega| \leq 0.90$ (transversality holds: $|\sin\omega| \geq 0.45$), and within-unit residual correlation $\rho_{SY} = 0.524$. Table 4 traces the corner mechanism of Section 3.2 across 300 replications per sample size (sieve first stage, fixed K).

Table 4: Corner anatomy on the KRR-calibrated oracle (300 reps, fixed sieve dimension). "Total" is the implied plug-in corner bias $\sum_{x^*} \mathbb{E}[\Delta_g \Delta_h]\, w/(\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|)$ in θ units; "SS estimate" is the mean corner cross correction $\widehat{\text{corner}}$ of (23), which by split-sample independence targets the *stochastic* (K/n) part of the diagonal; "remainder" is their difference, the bias-product ($K^{-2s/d}$) part. Sign convention: the anatomy script measures $\mathbb{E}[\Delta\hat{\tau}_S\,\Delta\hat{\tau}_Y]$ (positive here, as $\rho_{SY} > 0$); the table reports the corner term in θ units, where the harm-share convention $h_0 = -\tau_Y$ flips its sign. $\overline{\text{corr}}$ averages $\text{Corr}(\Delta_g(x^*), \Delta_h(x^*))$ over the ten corners. The column implements the corner *cross* term $\mathcal{B}_{\mathcal{C}}^{\times}$ of Proposition 4.7 — the first term of the bracket in (17) — which by (24) is exactly the quantity that margins-only debiasing leaves behind and that $\widehat{\text{corner}}$ targets, so the comparison across columns is like-for-like. It is *not* the corner's full contribution to the plug-in bias, which also contains the two $\cos\omega$-weighted diagonals of (17) (non-negligible here, since $|\cos\omega|$ reaches 0.90) but which margins-only debiasing already removes. The column also uses an unnormalized covariate density (Section 6.4, item (L2)).

n	Total corner bias	SS estimate	Remainder	$\overline{\text{corr}}(\Delta_g, \Delta_h)$	ρ_{SY}
2000	−0.0080	−0.0044	−0.0036	0.50	0.524
4000	−0.0048	−0.0022	−0.0025	0.46	0.524
8000	−0.0038	−0.0012	−0.0026	0.45	0.524

Three predictions of Section 3.2 are consistent with the table. *(i)* The pointwise correlation of the two surface errors at the corner tracks the within-unit residual correlation (0.45–0.50 against $\rho_{SY} = 0.524$), which is the mechanism behind the V_{gh} term in (17). *(ii)* The stochastic part of the corner diagonal halves as n doubles at fixed K ($-0.0044 \to -0.0022 \to -0.0012$) — the K/n scaling — and is tracked by the SS corner correction, which by construction cancels bias parts across the two independent halves. *(iii)* The remainder is roughly constant in n at fixed K (-0.0036 at $n = 2000$, then ≈ -0.0025), as befits the $K^{-2s/d}$ bias product, which is controlled by undersmoothing K rather than by debiasing — and which, per Proposition 4.7, the SS statistic is not supposed to capture. The measured corner bias (0.4–0.8 percentage points of a $\theta_0 = 0.161$ target) is material relative to the plug-in standard error, and its sign matches the Monte Carlo finding that the corner-aware and margins-only SS estimates differ by $+0.004$ to $+0.005$ at $n = 2000$. This validates the cross-corner component of (17) — the component that Proposition 4.7 says is at stake in the margins-only comparison. The two

$\cos\omega$ components of (17) are not tested here, because both debiasing variants remove them; testing them would require comparing the *plug-in* against a corner-corrected plug-in.

6.3 Results

Table 2 and Figures 1–2 report all cells; Section 6.4 states which conclusions they can and cannot support.

(1) Smooth truth (KRR): the plug-in is already good; corner-aware SS removes what bias there is. The plug-in covers at 0.948–0.962 at all n with a small decaying bias ($+0.010 \to +0.003$) and a well-calibrated standard error (SE/SD 0.94–1.11, tightening toward one with n). SS debiasing removes the bias essentially completely ($+0.003 \to +0.0007$; a 3.5-fold bias reduction at $n = 1000$ and a tenfold reduction at $n = 2000$) at negligible variance cost by $n \geq 4000$. The *margins-only* correction overshoots to negative bias (-0.006 to -0.0005) at every n: it removes the (positive) margin diagonals — and, per (24), the two $\cos\omega$ corner diagonals as well — but not the (negative) corner *cross* term. The gap between the corner-aware and margins-only estimates equals the estimated corner correction ($\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\text{mar}} - \widehat{\text{corner}}$) and matches the anatomy of Table 4 to the fourth decimal ($+0.0045$ at $n = 2000$ against minus the anatomy's SS corner estimate of -0.0044). The raw corner cross form $q_{\text{full}} - q_g - q_h$ (eight times the corner correction) halves as n doubles ($-0.070 \to -0.036 \to -0.020 \to -0.0095$) at fixed K, the K/n scaling of (17). Coverage of the SS intervals is 0.894–0.934 on this design: an improvement in centering bought at a 2–5 point coverage cost relative to the plug-in, which we do not describe as honest coverage.

(2) Rough truth (scalar WGAN, $s \approx 1$): debiasing accelerates the bias decay. This design violates the smoothness hypothesis of every theorem above ($s \approx 1 < (d+1)/2 = 1.5$), so it is a robustness probe. The plug-in bias is large and decays at only $\approx n^{-0.34}$ ($+0.064 \to +0.032$), and its near-nominal coverage rests on a conservative SE (SE/SD 1.1–1.3). SS debiasing accelerates the empirical decay to $\approx n^{-0.55}$ ($+0.066 \to +0.021$) with SE/SD 0.83–0.98 and coverage $0.87 \to 0.93$. Because K is fixed across n here, these exponents describe the finite-sample behavior of the estimators at $K = 25$; they are not estimates of the asymptotic rate, which would require K_n to grow (Section 6.4, item (L4)). On this rough truth the margins-only variant is slightly better centered than the corner-aware one — with $s \approx 1$ the corner correction is itself noisily estimated — a useful caution: corner-awareness is a bias-*decomposition* necessity, but its sampled version adds noise when the surfaces are rough.

(3) Machine-learning first stages: the projected Riesz correction works where the learner is adequate; the quadratic correction is not a free lunch. On the smooth truth the GBR bias falls monotonically along the ladder cross-fit $\to$ +Riesz $\to$ +quadratic at every n (at $n = 4000$:

$+0.024 \to +0.015 \to -0.005$), consistent with the two corrections targeting distinct error terms. But the composition has two finite-sample costs: the SS-on-GBR quadratic correction inflates the dispersion (by a factor of two at $n = 1000$, 1.4 thereafter), and the first-order SE does not account for it (SE/SD ≈ 0.6), so DD intervals undercover (0.69–0.83); and at $n = 8000$ on KRR the quadratic correction overshoots (-0.011). This is what Remark 4.4 predicts: (21) is a fourth-order finite difference of a function that, for tree first stages, is a step function, so Theorem 5.5 simply does not apply to this cell. On the rough truth GBR is bias-limited ($+0.045$ nearly flat in n; coverage falls to 0.64): no first-order correction can rescue a nuisance that misses the kinks. The degenerate arm is the extreme case: GBR manufactures a spurious harm share of $+0.17$ (coverage 0), and there the quadratic correction *does* rescue it ($+0.007$, coverage 0.87 at $n = 8000$).

(4) The two-band SE is well calibrated for the sieve family. The variance estimator (8) includes the empirical-measure term $\hat{\varsigma}^2/n = \hat{\theta}(1 - \hat{\theta})/n$ — the sampling error of the average given the surfaces, the analogue of the $\mathrm{Var}([h_0]_+)$ term the single-threshold unknown-density welfare theorem adds — which is 6–12% of the SE at these sample sizes and, if omitted, produces a spurious SE/SD of 0.87–0.95. With it included, SE/SD is 0.90–1.11 for plug-in and SS on the KRR and affine designs, and the studentized intervals reach 0.93–0.95 by $n = 8000$; on the degenerate arm ($\theta_0 = 0$, empty margins) the sieve family degrades gracefully.

(5) Machine-learning first stages: why the sieve-DML interval undercovers, and when it does not. The cross-fitted GBR estimators cover only 0.58–0.88 on the smooth design, in sharp contrast to the sieve plug-in on the *same samples* — so the shortfall is a property of the first stage, not the DGP or the variance formula. The diagnosis is Remark 5.4. Measured against the oracle arm means, the R^2 of the boosted first-stage error on the tensor B-spline representer span is 0.21–0.34 across the four arm regressions (mean 0.26); the projected correction removes 31% of the plug-in's boundary bias. The direct variance test reported in Remark 5.4 shows that the remaining 69% is *bias*, not an independent added variance: the leading explanation is that the correction cannot see the part of the learner's error that lives on the margins, so it leaves the boundary displaced. Three levers move coverage: *(i) folds* — $K = 2 \to 5 \to 10$ lifts GBR coverage $0.76 \to 0.90 \to 0.91$ by shrinking the own-observation bias trained on $(K-1)/K$ of the sample; *(ii) representer basis* — matching a 400-feature random-feature representer to the boosted learner raises the projected R^2 from 0.26 to 0.53 and coverage from 0.90 to 0.94; *(iii) learner smoothness* — a random-feature (RBF) ridge, whose error the representer *can* span ($R^2 = 0.84$), attenuates the CATE by only about 1% (against boosting's 13%) and has a plug-in bias of $+0.0006$ (against $+0.016$), so there is almost nothing left to correct and, with a matched representer, it too reaches 0.94. Higher-capacity boosting is counterproductive — it sharpens the axis-aligned staircase, driving coverage

toward zero. The pattern reproduces the parent draft's own single-threshold Table 7. The residual 0.01–0.02 shortfall of even the best machine-learning cell below the sieve plug-in's coverage is within two Monte Carlo standard errors and should not be over-interpreted.

6.4 What the current Monte Carlo does and does not establish

The tables above are reproduced from v1 without change. Auditing the code against the theory turned up six implementation gaps. None of them affects Theorem 3.1 or the order accounting of Section 3.2, which are analytic; all of them bear on how much weight the simulation evidence can carry, and all are scheduled for repair before the next revision. We list them explicitly rather than leave them implicit.

(L1) The KRR "oracle truth" is displaced by uncentered residual resampling.
The sampler draws outcome noise by resampling the fitted per-arm KRR residual pools, which are not centered (ridge shrinkage leaves arm-specific residual means). The generated conditional means therefore differ from the nominal KRR surfaces by a constant per arm, shifting the generated CATEs by $\Delta\tau_S = +0.126$ and $\Delta\tau_Y = +0.182$ (the latter includes the short→long coupling). The reported truth θ_0 moves from 0.16115 to 0.15998 — immaterial for the bias columns — but the ten corner locations, gradients, angles, and first-stage errors used in Table 4 are all computed against the *wrong* surfaces. Fix: center each arm's residual pool before resampling, or define the oracle truth to include the shifts.

(L2) The corner density is unnormalized.
Covariates are drawn from the KDE conditional on $X \in [-3, 3]^2$ by rejection sampling, but the density accessor returns the untruncated KDE. The KDE mass of the box is 0.807, so the corner integrals in Table 4 should be scaled by $1/0.807 \approx 1.24$ before any comparison with (17). Together with the omission of the $\cos\omega$ terms noted in the table caption, the "Total" column is an incomplete implementation of the theory it is meant to check.

(L3) The KRR design violates Assumption 1.5(d).
The KDE density is positive at the truncation boundary, and a direct scan of the four box edges finds twelve sign crossings of the two contrast surfaces (τ_S: three on $x_1 = -3$, two on $x_1 = +3$, three on $x_2 = -3$; τ_Y: two on $x_1 = -3$, two on $x_2 = +3$). The margins therefore terminate at the support edge, and by Remark 1.6 the correct second derivative for this DGP carries additional support-edge $\mathcal{H}^{d-2}$ terms that (10) omits. Fix: taper w near $\partial\mathcal{X}$, or shrink the box so that all margins close inside it.

(L4) The main Monte Carlo holds K fixed.

Every sample size uses two knot segments per coordinate, i.e. $K = 25$ basis functions per arm at every n. The design therefore does not implement $K_n \asymp n^{d/(2s+1)}$, and the estimated log-rate slopes reported in Section 6.3(2) are not estimates of the asymptotic rate. What the design *does* deliver cleanly is the K/n scaling of the corner diagonal at fixed K (Table 4), which is the prediction Section 3.2 is actually testing.

(L5) The variance bandwidth does not satisfy Proposition 2.3.

The implementation sets $\varepsilon = \delta \cdot \widehat{\mathrm{sd}}(\hat{\tau})$ with δ fixed and then widens ε by factors of 1.5 until at least five observations enter the band. Since $\widehat{\mathrm{sd}}(\hat{\tau})$ converges to a positive constant, $\varepsilon_n \not\to 0$, contradicting Proposition 2.3 and Assumption 5.1(b). On the degenerate arm the widening manufactures a nonempty derivative band where the population margin is empty. Fix: $\varepsilon_n = \delta_n \cdot \widehat{\mathrm{sd}}(\hat{\tau})$ with $\delta_n \to 0$ at a rate satisfying $n\varepsilon_n/K_n \to \infty$.

(L6) The coded Riesz correction is not fold-specific.

Both the cross-fitted DML estimator and the DD estimator build a single full-sample Gram matrix and band representer from pooled out-of-fold predictions, whereas (26) and Theorem 5.2 require an off-fold representer for each fold. In the implementation, observation i's outcome can influence the global band through the model trained on i's own fold, which breaks the conditional mean-zero step used in Remark 5.4. Fix: build $\hat{v}^*_{-\kappa}$ inside the fold loop.

Taken together: Theorem 3.1 is verified analytically (Appendix C); the K/n order of the corner diagonal and the sign and magnitude of the corner correction are supported by Tables 4 and 2 at $d = 2$ and fixed K, subject to (L1)–(L3); the asymptotic rate claims of Theorems 2.2, 4.5, 5.2 and 5.5 are *not* tested by the present design, because K does not grow with n; and the mechanism behind the machine-learning undercoverage is identified as residual boundary bias rather than unmodelled variance, by the direct variance test in Remark 5.4.

Practical guidance. For sieve first stages, use the corner-aware SS estimator (21) with the two-band SE: it is tuning-free and removes the second-order stochastic diagonal including the corner, at a 2–5 point coverage cost relative to the plug-in in our designs. For a machine-learning first stage — needed when d is too large for a tensor sieve — prefer a *smooth* learner and *match the representer basis to the learner's own feature map*, so the projected correction is near-exact; avoid tree ensembles, whose piecewise-constant error is largest precisely on the margins the representer must span, and use at least five folds for the nuisance cross-fit. (This is guidance about the number of *nuisance* folds and does not conflict with the two-half construction of $\hat{\Theta}_{SS}$ and $\hat{\Theta}_{DD}$: there, the two halves furnish the independent replicates that the polarization identity (20) requires, and $K = 2$ is not a cross-fitting choice but part of the definition of the quadratic correction. The two can be combined — K-fold nuisance cross-fitting inside each half — at the cost of one more fit.) When a tree first stage is unavoidable, pair it with a full-refit bootstrap rather than the analytic SE. The comparison of the corner-aware and margins-only corrections is a useful specification diagnostic in its own right: their gap estimates the corner cross term, whose magnitude flags whether the two-threshold structure is quantitatively binding.

A Derivation of the Second Derivative (Theorem 3.1)

Write $T_g(t) := \int_{\{g_t=0\}\cap\{h_t\geq 0\}} \varpi_t \, \mathrm{d}\mathcal{H}^{d-1}$ with weight $\varpi_t := u\, w/\left\|\nabla g_t\right\|$ (we reserve $\omega(x)$ for the transversality angle throughout), the first term of (5) along γ_t; the second term is symmetric. Let $\mathbf{X}(\cdot, t)$ be the normal diffeomorphism carrying $\mathcal{M} := \{g_0 = 0\}$ to $\{g_t = 0\}$ with velocity $\dot{\mathbf{X}} = -(u/\left\|\nabla g_0\right\|)\, n_g$, and pull the integral back to $\mathcal{M}$:

$$T_g(t) = \int_{\mathcal{M}} \mathbb{1}\{h_t(\mathbf{X}(x,t)) \geq 0\}\; \varpi_t(\mathbf{X}(x,t))\; J_{\mathbf{X}}(x,t)\, \mathrm{d}\mathcal{H}^{d-1}(x).$$

Differentiating at $t = 0$ yields four contributions. *(1) Integrand:* $\partial_t \varpi_t = -u\, w\,(n_g \cdot \nabla u)/\left\|\nabla g_0\right\|^2$. *(2) Transport of ϖ and area:* by the surface-transport formula (Delfour and Zolésio, 2001, Ch. 9), $\partial_t\{\varpi(\mathbf{X})J_{\mathbf{X}}\} = -\big[(u/\left\|\nabla g_0\right\|)\, n_g \cdot \nabla\varpi + \varpi\,(u/\left\|\nabla g_0\right\|) H_g\big]$, the moving-level-set terms of Chen and Gao (2026, Lem. PathD–Level) with weight ϖ. *(3) Direct motion of the cut:* the indicator $\mathbb{1}\{h_t \geq 0\}$ evaluated on the (fixed, $t = 0$) manifold defines the region $\{h_0 + tv \geq 0\} \cap \mathcal{M}$; by the within-manifold Leibniz rule its derivative is an integral over the relative boundary $\mathcal{C}$ with normal velocity $v/\left\|\nabla_{\mathcal{M}} h_0\right\|$, where $\left\|\nabla_{\mathcal{M}} h_0\right\| = \left\|P_T \nabla h_0\right\| = \left\|\nabla h_0\right\| |\sin\omega|$, giving $+\int_{\mathcal{C}} \varpi_0\, v/(\left\|\nabla h_0\right\| |\sin\omega|)\, \mathrm{d}\mathcal{H}^{d-2}$ — the cross term. *(4) Induced motion of the cut:* the composition $h_0(\mathbf{X}(x,t)) = h_0(x) - t\,(u/\left\|\nabla g_0\right\|)\, n_g \cdot \nabla h_0 + o(t) = h_0(x) - t\, u\, \left\|\nabla h_0\right\| \cos\omega/\left\|\nabla g_0\right\| + o(t)$ perturbs the cut function by an effective $v_{\text{eff}} = -u\left\|\nabla h_0\right\| \cos\omega/\left\|\nabla g_0\right\|$; the same within-manifold Leibniz rule then contributes $-\int_{\mathcal{C}} \varpi_0\, u\cos\omega/(\left\|\nabla g_0\right\| |\sin\omega|)\, \mathrm{d}\mathcal{H}^{d-2}$ — the diagonal corner term of (11). Collecting (1)–(4), and adding the symmetric derivative of the $\mathcal{M}_h$-term (which contributes the mirror-image margin form, the second copy of the cross term, and the v^2 diagonal corner term), gives (10)–(11). Transversality bounds every corner integrand; twice-differentiability of g_0, h_0 and Assumption 1.4(d) bound the curvature and gradient terms. $\square$

B Orders of the Remainder Terms

The margin components of all remainder bounds are adaptations of the single-threshold arguments — Chen and Gao (2026, Thms. 5, 8 and the SS/LOO constructions) for the quadratic terms, and the extended single-threshold draft (sieve-DML theorem and its Assumption 5) for the cross-fitted first-order correction — applied margin by margin with the truncation indicators carried as bounded weights. This appendix records the two order calculations that are specific to the two-threshold geometry and that v1 got wrong.

B.1 Why the margin diagonal is $K_n^{1+1/d}/n$

Substituting the stochastic part (12) into Q_g and separating $i = j$ from $i \neq j$,

$$Q_g[u_g, u_g] = \underbrace{\frac{1}{n^2}\sum_i \epsilon_{S,i}^2\, Q_g[s_i, s_i]}_{T_{\text{diag}}} + \underbrace{\frac{1}{n^2}\sum_{i\neq j} \epsilon_{S,i}\epsilon_{S,j}\, Q_g[s_i, s_j]}_{T_{\text{off}}}.$$

The leading integrand of Q_g carries one gradient, so $\mathbb{E}\, Q_g[s_i, s_i]$ involves $\mathbb{E}[s_i(x)\, n_g \cdot \nabla s_i(x)] = b(x)' G^{-1} \nabla b(x) \cdot n_g \asymp K_n^{1+1/d}$ by (13), whence $\mathbb{E}\, T_{\text{diag}} \asymp n^{-2} \cdot n \cdot K_n^{1+1/d} = K_n^{1+1/d}/n$. This is the order stated by Chen and Gao (2026) for the single-threshold value functional, and it is the order that Table 1 records. The curvature part of Q_g, which carries no gradient, contributes only K_n/n.

The corner forms in (10)–(11) carry *no* gradient of u or v. Their diagonals therefore involve $\mathbb{E}[s_i(x)^2] = b(x)' G^{-1} b(x) = \mathcal{V}(x) \asymp K_n$ and are K_n/n: bounded by $K_n^{-1/d}$ times the margin bound. v1's assertion that the corner diagonal is "of the same order K_n/n as the margin diagonals" conflated a bound with an exact order.

Two asymmetries between the two statements should be recorded. *(i)* The corner order is exact: $\mathcal{V} > 0$ and enters undifferentiated, so integrating it over $\mathcal{C}$ against a weight of constant sign cannot cancel. *(ii)* The margin order is a bound only. $\mathbb{E}[s_i\, \partial_n s_i] = \frac{1}{2}\partial_n \mathcal{V}$ under homoskedasticity, and $\mathcal{V}$ oscillates at the mesh scale with amplitude $\asymp K_n$, so while $\|\nabla \mathcal{V}\|_\infty \asymp K_n^{1+1/d}$ pointwise, the surface integral in (18) can cancel. Expanding $\mathcal{V} = \bar{\mathcal{V}} + \tilde{\mathcal{V}}$ with $\tilde{\mathcal{V}}$ mesh-periodic and mean zero, and writing $\tilde{\mathcal{V}}(x) = \sum_{j\neq 0} c_j e^{2\pi i j \varkappa^{-1} x}$, the integral of $\partial_n \mathcal{V}$ along a margin of slope α (in $d = 2$) picks up a non-decaying contribution only from pairs (j, l) with $j + l\alpha \approx 0$, i.e. only if α is rational with small denominator; for cubic splines $|c_j| = O(j^{-4})$, so even moderate denominators are heavily suppressed. Appendix C, item 5, confirms this: $\alpha \in \{0, 1\}$ give $\asymp K_n^{1+1/d}$, $\alpha \in \{1/2, 8/5\}$ and irrational α give $\asymp K_n$. Consequently the margin bound is attained for mesh-resonant margins — in particular for $d = 1$, where $\mathcal{M}_g$ is a point and no averaging occurs, so the single-threshold rate of Chen and Gao (2026) is unaffected — but not necessarily otherwise.

B.2 Off-diagonal (cross-half) orders

T_{off} is a degenerate second-order U-statistic. Writing $a_i := G^{-1}b(X_i)$, every quadratic form in (10) evaluates to $a_i' M a_j$ for a symmetric matrix M built from the relevant boundary integral, so $\mathrm{Var}(T_{\text{off}}) \asymp n^{-2}\,\mathrm{tr}\{(MG^{-1})^2\}$. For tensor B-splines with mesh $\varkappa = K_n^{-1/d}$, a normalized basis function has height $\asymp \varkappa^{-d/2}$ on a cell of volume $\varkappa^d$, so for a p-dimensional integration set $\mathcal{S}$ and ℓ gradient factors,

$$\#\{k : \mathrm{supp}(b_k) \cap \mathcal{S} \neq \emptyset\} \asymp \varkappa^{-p}, \qquad M_{kk} \asymp \varkappa^{-d}\varkappa^{p}\varkappa^{-\ell} = \varkappa^{p-d-\ell},$$

and therefore $\mathrm{tr}(M^2) \asymp \varkappa^{-p}\varkappa^{2(p-d-\ell)} = \varkappa^{p-2d-2\ell} = K_n^{(2d+2\ell-p)/d}$. Substituting:

Term	(p,ℓ)	$\mathrm{tr}(M^2)$	T_{off}
margin, gradient part	$(d-1,1)$	$K_n^{(d+3)/d}$	$n^{-1}K_n^{(d+3)/(2d)}$
corner	$(d-2,0)$	$K_n^{(d+2)/d}$	$n^{-1}K_n^{(d+2)/(2d)}$
margin, curvature part	$(d-1,0)$	$K_n^{(d+1)/d}$	$n^{-1}K_n^{(d+1)/(2d)}$

The first line reproduces the $n^{-1}\sqrt{K_n^{(d+3)/d}}$ bound of Chen and Gao (2026), which is a useful check on the calculus. Two consequences. First, all three are $O(r_n^*)$ at $K_n \asymp n^{d/(2s+1)}$ under $s \geq (d+1)/2$ (the corner needs only $s \geq d/2$), so the off-diagonals never bind before the terms in Table 1's third block do. Second, and correcting v1: it is *not* true in general that a lower-dimensional integration set gives a smaller cross-half term. Reducing p by one *raises* $\mathrm{tr}(M^2)$ by $K_n^{1/d}$, because there is less averaging, not more. Comparing like with like, the codimension-two corner has a *larger* generic off-diagonal than a hypersurface with the same integrand. The corner off-diagonal ends up smaller than the leading margin off-diagonal only because the margin form carries a gradient factor ($\ell = 1$) that costs $K_n^{2/d}$ in $\mathrm{tr}(M^2)$, which more than offsets the corner's $K_n^{1/d}$ dimension gain; it remains *larger* than the margin's curvature part.

B.3 Status of the proofs

The joint CLT for the two-band linear term follows from the Cramér–Wold device and the Lindeberg–Lyapunov condition of Assumption 5.1(g) applied to the summed per-unit influence $v_S^*(X_i,W_i)\epsilon_{S,i} + v_Y^*(X_i,W_i)\epsilon_{Y,i}$; the within-unit correlation ρ_{SY} is absorbed into σ_n^2 by construction (8). Beyond the order calculations of B.1 and B.2, complete proofs of

Theorems 4.5, 5.2 and 5.5 are not yet written; the statements above are formulated so that the cited single-threshold lemmas apply to the margin parts verbatim, and the corner parts require only the transversality bound and the dimension count $\dim \mathcal{C} = d-2$. The two places where this is not merely mechanical are (i) the uniform control of the estimated geometry along the jackknife path (Assumption 4.1), which is an assumption rather than a result, and (ii) the moving-boundary remainder Assumption 5.1(f), whose primitive sufficient conditions for generic machine-learning first stages we do not currently have.

C Numerical Verification of Theorem 3.1

Theorem 3.1 was checked against exact area computations in $d = 2$ on four designs, chosen so that between them every term of (10)–(11) is activated in isolation. In each case $\Theta(t) = \int \mathbb{1}\{g_0 + tu \geq 0\}\mathbb{1}\{h_0 + tv \geq 0\}w$ has a closed form; $\Theta''(0)$ is obtained by a Richardson-extrapolated central second difference and compared with the right-hand side of (10).

1. *Radially weighted disk.* $g_0 = 1 - \|x\|^2$, no second threshold, $w = \rho(\|x\|)$, $u \equiv a$. Only the transport and curvature terms of Q_g are active, and (11) reduces to $\pi a^2 \rho'(1)/2$. Relative error 6×10^{-10}.

2. *Disk with a non-constant direction.* $g_0 = 1 - \|x\|^2$, $w \equiv 1$, $u(x) = \|x\|^2$, so $\{g_t \geq 0\}$ is the disk of radius $(1-t)^{-1/2}$ and $\Theta(t) = \pi/(1-t)$. This activates the $u\,(n_g \cdot \nabla u)$ term. Formula and truth both give $D\Theta = \pi$, $D^2\Theta = 2\pi$ to ten digits.

3. *Disk cut by an oblique chord.* $g_0 = 1 - \|x\|^2$, $h_0 = x_1 - \kappa$, $w \equiv 1$, $u \equiv a$, $v \equiv c$. Both margin forms vanish, isolating the two corner diagonals and the corner cross term at $\cos\omega = -\kappa$. Checked at $\kappa \in \{0.6, 0.2, -0.5\}$ (both signs of $\cos\omega$) and $(a, c) \in \{(1, 1), (1.7, -0.9), (0.4, 2.1)\}$; maximum relative error 2×10^{-10}.

4. *Lens of two disks.* $g_0 = 1 - \|x - p\|^2$, $h_0 = 1 - \|x - q\|^2$ with $\|p - q\| = D$, $w \equiv 1$, $u \equiv a$, $v \equiv c$; both surfaces curved and obliquely intersecting, with $\cos\omega = 1 - D^2/2$. The prediction $\{ac - \frac{1}{2}(a^2 + c^2)\cos\omega\}/|\sin\omega|$ matches the exact lens area's second derivative at $D \in \{0.8, 1.2, 1.7\}$ and the same three (a, c) pairs; maximum relative error 2×10^{-9}.

Designs 3 and 4 are the substantive ones. Varying (a, c) over three non-proportional pairs identifies the quadratic form in (a, c) separately, so they confirm all three coefficients

$$\underbrace{\frac{2}{\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|}}_{\text{cross}}, \qquad \underbrace{\frac{-\cos\omega}{\|\nabla g_0\|^2\,|\sin\omega|}}_{Q_g \text{ diag.}}, \qquad \underbrace{\frac{-\cos\omega}{\|\nabla h_0\|^2\,|\sin\omega|}}_{Q_h \text{ diag.}},$$

and their relative signs, at both signs of $\cos\omega$.

The sharpest single check is design 4 with a *one-sided* perturbation, $v \equiv 0$: there the cross term is identically zero, so anything nonzero must come from the Q_g corner diagonal. At $D = 0.8, 1.2, 1.7$ ($\cos\omega = +0.680, +0.280, -0.445$) and $a = 1$ the exact lens curvatures are -0.463713017, -0.145833333, $+0.248456065$, against formula values -0.463713017, -0.145833333, $+0.248456064$ — including the sign reversal as $\cos\omega$ crosses zero. This is the term that v1's order discussion omitted and that (17) restores: the corner contributes to the plug-in bias even when the two contrast errors are uncorrelated, provided the margins do not meet orthogonally.

5. Polarization and mesh resonance. Two further checks support statements made in the body.

(a) The polarization identity (24). On all thirteen (D or κ, a, c) configurations of designs 3 and 4, the numerically differenced $q_{\text{full}} - q_g - q_h$ agrees with $2C[a, c]$ to full double precision, while the sum of the two $\cos\omega$ diagonals — which is nonzero and often larger in magnitude — is absorbed entirely into q_g and q_h. For instance at $D = 1.2$, $a = c = 1$: $q_{\text{full}} = 0.750000$, $q_g = q_h = -0.145833$, so $q_{\text{full}} - q_g - q_h = 1.041667 = ac/|\sin\omega| = 2C$, against a corner-diagonal sum of -0.291667. This is what licenses Proposition 4.7: margins-only debiasing removes the $\cos\omega$ diagonals and leaves the cross term.

(b) Mesh resonance in (18). With periodic uniform cubic tensor B-splines on the torus in $d = 2$ ($K_n = m^2$, mesh $\varkappa = 1/m$), we computed $\mathcal{V} = \mathcal{V}_1 \otimes \mathcal{V}_1$ in closed form — the diagnostic $\int_0^1 \mathcal{V}_1 = m$ holds exactly and $\|\mathcal{V}_1'\|_\infty = 1.226\, m^2$ to four digits for $m = 16, 32, 64$, confirming the pointwise scalings (13) — and evaluated $\int_{\mathcal{M}_g} \psi\, \partial_n \mathcal{V}$ along straight margins of slope α for $m = 16, 32, 64, 128$. Reporting $|I|/K_n$ (flat $\Rightarrow I \asymp K_n$; growing linearly in $m \Rightarrow I \asymp K_n^{1+1/d}$):

α	$m = 16$	32	64	128
0 (mesh-aligned)	16.1	20.7	66.6	27.4
1 (resonant)	3.6	5.2	15.4	6.9
$1/2$, $8/5$ (rational, $q \geq 2$)	≤ 0.12	≤ 0.20	≤ 0.09	≤ 0.59
$1/\pi$, $\sqrt{2}$, $\sqrt{5} - 1$ (irrational)	≤ 0.43	≤ 0.65	≤ 0.40	≤ 0.48

The first two rows fluctuate in phase but grow with m (so $I \asymp K_n^{1+1/d}$); the last two are bounded (so $I \asymp K_n$). The resulting hundredfold spread at $m = 128$ is the evidence for Remark 3.3.

6. The support-edge term of Remark 1.6. Take the unit disk $g_0 = 1 - \|x\|^2$ with the *fixed* support $\mathcal{X} = \{x_1 \geq \kappa\}$, $w \equiv 1$, $u \equiv a$, and no second threshold. The exact curvature of the circular-segment area matches $-2a^2 \cos\omega_\partial/(4\,|\sin\omega_\partial|)$ with $\cos\omega_\partial = -\kappa$ at

$\kappa \in \{0.6, 0.2, 0, -0.5\}$ and $a \in \{1, 1.7\}$ to ten digits, including the exact vanishing at $\kappa = 0$ (orthogonal meeting). This confirms both the form of the edge term and that it carries no cross component.

The scripts for items 1–6 are `verify_D2_theorem.py`, `verify_polarization.py` and `verify_mesh_resonance.py`, alongside this document.

References

Athey, S., R. Chetty, G. W. Imbens, and H. Kang (2024): "The Surrogate Index: Combining Short-Term Proxies to Estimate Long-Term Treatment Effects More Rapidly and Precisely," *Working paper*.

Athey, S., G. W. Imbens, J. Metzger, and E. Munro (2021): "Using Wasserstein Generative Adversarial Networks for the Design of Monte Carlo Simulations," *Journal of Econometrics*, forthcoming.

Banerjee, A., E. Duflo, N. Goldberg, D. Karlan, R. Osei, W. Pariente, J. Shapiro, B. Thuysbaert, and C. Udry (2015): "A Multifaceted Program Causes Lasting Progress for the Very Poor: Evidence from Six Countries," *Science*, 348(6236), 1260799.

Chen, X., and W. Y. Gao (2026): "Thin Sets Are Not Equally Thin: Minimax Learning of Submanifold Integrals," *Working paper*; first version arXiv:2507.12673v1.

Chen, X., Z. Chen, and W. Y. Gao (2025): "Inference on Policy Value and Welfare under Optimal Treatment Rules," *Working paper*, and its extended draft (quadratic debiasing; sieve DML).

Chen, J., and D. M. Ritzwoller (2023): "Semiparametric Estimation of Long-Term Treatment Effects," *Journal of Econometrics*, 237(2), 105545.

Delfour, M. C., and J.-P. Zolésio (2001): *Shapes and Geometries: Analysis, Differential Calculus, and Optimization*. SIAM, Philadelphia.

Evans, L. C., and R. F. Gariepy (2015): *Measure Theory and Fine Properties of Functions*, revised ed. CRC Press, Boca Raton.

Viviano, D., and J. Bradic (2024): "Fair Policy Targeting," *Journal of the American Statistical Association*, 119(545), 730–743.

Whitehouse, J., Q. Chen, M. Austern, and V. Syrgkanis (2026): "Inference on Optimal Policy Values and Other Irregular Functionals via Softmax Smoothing," arXiv:2507.11780v2.